Generative Modelling for Realistic Mars Landing Site Visualisation

Deepakraj Rajmohan

Master of Science in Artificial Intelligence

from the

University of Surrey

School of Computer Science and Electrical and Electronic Engineering

Faculty of Engineering and Physical Sciences

University of Surrey

Guildford, Surrey, GU2 7XH, UK

September 2026

Supervised by: Dr Xiatian Zhu

DECLARATION OF ORIGINALITY

I confirm that the project dissertation I am submitting is entirely my own work and that any material used from other sources has been clearly identified and properly acknowledged and referenced. In submitting this final version of my report to the JISC anti-plagiarism software resource, I confirm that my work does not contravene the university regulations on plagiarism as described in the Student Handbook. In so doing I also acknowledge that I may be held to account for any particular instances of uncited work detected by the JISC anti-plagiarism software, or as may be found by the project examiner or project organiser. I also understand that if an allegation of plagiarism is upheld via an Academic Misconduct Hearing, then I may forfeit any credit for this module or a more severe penalty may be agreed.

"Generative Modelling for Realistic Mars Landing Site Visualisation"

Deepakraj Rajmohan

Author Signature Date:

Supervisor's name: Dr Xiatian Zhu

WORD COUNT

Number of Pages: 61

Number of Words: approximately 13,000 (main text, Chapters 1–7, excluding front matter, tables, and bibliography)

ABSTRACT

Mission planning and astronaut/operator training for planetary exploration increasingly rely on immersive VR reconstructions of candidate landing sites, but the only data available for most such sites is nadir orbital imagery (e.g. NASA's HiRISE, 25 cm/px), not the ground-level, perspective views that make a VR scene feel navigable. This dissertation addresses that nadir-to-perspective domain gap in the context of ESA's ExoMars Rosalind Franklin rover mission, proposing a three-stage generative pipeline — diffusion-based super-resolution, physics-constrained unpaired view translation, and Gaussian Splatting scene reconstruction — to turn orbital imagery into explorable, physically plausible VR environments.

The literature review and methodology survey the super-resolution, image-to-image translation, cross-view synthesis, neural rendering, and hallucination-safety literature relevant to each stage, and formalise a Neural Schrödinger Bridge formulation, evaluation framework (FID, shadow-angle error, hallucination rate, VR frame rate), and baseline comparisons (MARSGAN, DiffusionSat, CycleGAN, UNSB-no-physics) as the proposed architecture. A CycleGAN baseline for Stage 2 was implemented, trained on a corrected real-resolution HiRISE corpus (after an initial corpus was found to use browse-quality, ˜3 m/px imagery rather than the true 25 cm/px RDR products), and evaluated at FID 269.9, worse than the buggy browse-resolution corpus's 171.8. Diagnosis traced this to two causes: an epoch-3 discriminator shortcut, and a deeper corpus-diversity collapse (seven observations in one region, rather than the intended ˜300 across four) discovered by auditing the corrected corpus itself. A literature review conducted in response identified a cross-view synthesis cluster in Earth remote sensing that solves this exact translation direction via an explicit geometric intermediate rather than pixel-space translation, motivating a redesign: a loss-tuned CycleGAN retrain that isolates the discriminator-shortcut fix (FID 95.8, the best CycleGAN result, though still short of the ≤55 target and architecturally unable to translate viewpoint); a geometry-mediated corpus built from 8 real HiRISE stereo DTMs and a deterministic ground-view renderer, which does translate viewpoint and breaks the shortcut but shows compressed texture diversity; a single-image DTM estimator (RMSE 2.02 m against real Gale Crater ground truth) built to extend geometry-mediation beyond the 8 sites with real stereo coverage; a pipeline recovering genuine paired supervision from real rover pose metadata; and a ControlNet texture-synthesis stage, trained on that paired data and chained end-to-end after the renderer, which produced the

first clean evidence that the DTM estimator's height error visibly affects the pipeline's final output rather than being masked by an under-textured test case. A kernel-inception-distance metric, valid regardless of sample size, is reported across all three generative stages for a direct comparison FID alone cannot give.

The geometry-mediated pipeline that emerged from this diagnosis is architecturally different from the physics-constrained Neural Schrödinger Bridge originally proposed; Chapter 3 states the evidence-based rationale for that change in full, and Chapter 6 records how it, together with the project's other open scope questions (Stage 1 super-resolution, Stage 3 Gaussian Splatting, and Track 2's corpus-diversity limitation), was resolved for this submission. Every quantitative target the original methodology set is reported against what was actually measured (Chapter 5), and Chapter 7 assesses the work against its four original objectives directly, including where it fell short and why.

CONTENTS

LIST OF FIGURES

1 INTRODUCTION

Planetary exploration missions increasingly depend on immersive virtual-reality (VR) environments for mission planning, hazard assessment, and astronaut or rover-operator training. Building a convincing VR reconstruction of a Mars landing site, however, runs into a basic data problem: the only imagery available for almost every candidate site, including all ExoMars candidate landing areas, is nadir orbital imagery captured from orbit looking straight down, whereas a usable VR scene needs ground-level, perspective imagery of the kind a rover's own cameras produce after it has landed. This dissertation is concerned with closing that nadir-to-perspective domain gap using generative modelling, so that orbital data alone — available years before any rover touches down — can be turned into a physically plausible, explorable VR surface.

The work is computational and experimental in nature: it combines remote-sensing data engineering (acquiring and correctly processing real orbital and rover imagery), generative model design and training (image-to-image translation and super-resolution networks), and, in later stages, real-time neural rendering. The concrete application context is ESA's ExoMars Rosalind Franklin rover mission, and the primary test site is Oxia Planum, the mission's selected landing area.

1.1 Background and Context

High-resolution orbital imagery of Mars is comparatively abundant: NASA's HiRISE instrument has imaged large fractions of the surface at up to 25 cm/pixel, resolving individual boulders and small craters. This is nadir imagery — captured looking straight down from orbit — and it is systematically unlike the perspective, oblique, ground-level imagery captured by rover cameras such as NAVCAM, which show terrain from roughly a metre above the surface, with strong foreshortening, occlusion, and local lighting effects that a top-down orbital image simply does not contain. No dataset of geographically *aligned* nadir/perspective Mars image pairs exists, because rover imagery only exists for the handful of sites a rover has actually visited, and even then the rover was never positioned to recreate an orbital viewing geometry. Any method that aims to synthesise rover-perspective imagery from orbital data must therefore work without paired supervision.

This gap sits at the intersection of several active research areas surveyed in Chapter 2: single-image super-resolution (to recover resolution lost between coarser orbital instruments and HiRISE), unpaired image-to-image translation (CycleGAN-style methods, and more recent physics-informed variants such as the Neural Schrödinger Bridge), neural scene rendering (NeRF and Gaussian Splatting for turning a set of synthetic views into an explorable 3D scene), and the emerging concern, specific to generative models used for scientific and mission-critical purposes, of hallucination: a model may produce a plausible-looking rock or crater that was never actually present, which is a materially different failure mode from a blurry or low-fidelity output. Existing single-modality approaches (pure super-resolution, or pure style-transfer GANs) each solve part of this problem but none combines resolution recovery, physically-grounded view translation, and real-time rendering into a single pipeline validated against Mars-specific physical plausibility criteria, which is the gap this dissertation addresses.

1.2 Objectives

The objectives below were carried forward from the initial project proposal and refined during the methodology stage (Chapter 3) once the literature review had established that no existing method closes the nadir-to-rover-perspective gap end-to-end for Mars surface synthesis:

- **O1.** Produce nadir imagery at 25 cm/pixel from coarser CaSSIS/CTX inputs when HiRISE coverage is absent.
- **O2.** Translate unpaired HiRISE nadir patches into statistically plausible rover-perspective images whose photometric and morphometric properties are consistent with Mars surface physics.
- **O3.** Reconstruct a real-time renderable 3D scene from the synthetic perspective images at VR frame rates (≥ 72 fps).
- **O4.** Quantify the fidelity and physical plausibility of generated imagery against held-out real rover data and known planetary surface statistics.

Effort concentrated on the data engineering and modelling required to make O2 tractable: building correct, real-resolution training corpora for both the orbital and rover domains, implementing and rigorously evaluating a CycleGAN baseline, and, once that baseline's diagnosis pointed at a specific architectural cause, designing and evaluating the geometry-mediated pipeline that replaced the physics-constrained bridge originally proposed for O2 (Chapter 3, Section 3.2.3

states why). O1 and O3 were formally scoped out of this submission (Chapter 6, Sections 6.3–6.4); O4 is addressed partially, against FID, KID, and RMSE, but not the shadow-angle, Hapke, and hallucination-rate metrics Chapter 3 also specified. Chapter 7 assesses all four objectives against this final scope in full.

1.3 Achievements

The following was completed:

- A full literature review (Chapter 2) covering eight topic clusters — orbital data constraints, super-resolution, image-to-image translation, cross-view synthesis, digital terrain modelling, neural scene rendering, VR reconstruction, and generative hallucination — synthesising 23 papers into a comparative analysis that motivates the proposed architecture, extended after the original CycleGAN baseline's failure with a cluster of Earth-remote-sensing cross-view synthesis work absent from the initial review.
- A complete methodology (Chapter 3) formalising the three-stage pipeline as originally proposed, including the Neural Schrödinger Bridge formulation with four physics-informed loss terms, a full evaluation protocol against four baselines, and, added once the project's own evidence diverged from that proposal, a documented, reasoned account of the architecture actually implemented and why (Section 3.2.3).
- A real-resolution data pipeline (Chapter 4) that corrected a data-quality bug in the original HiRISE corpus (browse-quality ~3 m/px imagery masquerading as the intended 25 cm/px product) and rebuilt it from genuine RDR products under a tight disk-quota constraint on the training machine.
- A trained and evaluated CycleGAN baseline (Chapter 5) for the Stage 2 view-translation problem, including a specific, evidence-based two-level diagnosis of why its FID score (269.9) fell well short of the target (≤ 55): a discriminator collapse traced to the epoch level using the training logs, and, behind it, a corpus-diversity collapse found by auditing the corpus itself.
- A geometry-mediated redesign (Chapters 4–5) built in response to that diagnosis: a loss-tuned CycleGAN retrain isolating the discriminator fix (FID 95.8, best CycleGAN result), a real-DTM ground-view renderer and geometry-mediated corpus that does translate viewpoint, a single-image DTM estimator (RMSE 2.02 m) extending that corpus beyond the sites

with real stereo coverage, a real pose-matched pairing pipeline recovering genuine (condition, target) supervision from rover metadata, a paired ControlNet texture-synthesis stage trained on it, and an end-to-end chain giving the first clean evidence that the estimator's error visibly affects the final output.

- A common evaluation metric (KID) computed across all three generative stages despite their very different sample sizes, and a quantitative, rather than purely visual, check of the end-to-end pipeline's two chains (Chapter 5, Section 5.7).
- Explicit scope decisions, argued rather than left implicit, for every part of the original proposal not carried into the final system: Stage 1 super-resolution, Stage 3 Gaussian Splatting, Track 2's corpus-diversity limitation, and the UNSB-versus-geometry-mediated architecture question (Chapter 6).

1.4 Overview of Dissertation

Chapter 2 reviews the literature underpinning each stage of the proposed pipeline and identifies the specific gap this work fills. Chapter 3 presents the methodology: the mathematical formulation of the three-stage architecture as originally proposed, the architecture actually implemented, the reasoning behind that change, the dataset design, and the evaluation framework. Chapter 4 describes what was implemented: the data acquisition and preprocessing pipelines for both the HiRISE and rover corpora, the CycleGAN model and training infrastructure built to establish a Stage 2 baseline, and the geometry-mediated redesign (a real-DTM renderer, a single-image DTM estimator, a real pose-matched pairing pipeline, and a texture-synthesis ControlNet) built once that baseline's diagnosis pointed beyond a loss-tuning fix. Chapter 5 presents results: the original run's loss behaviour and FID, its two-level diagnosis, two further CycleGAN tracks that isolate the loss-weight question from the viewpoint-translation question, the DTM estimator's evaluation, an end-to-end chain test giving the first evidence that the estimator's error propagates to the final image, a common-metric comparison across all three generative stages, and every quantitative target from Chapter 3 measured against what was actually achieved. Chapter 6 records how the architecture reconciliation and every other scope question this project faced were resolved, and what remains as genuine future work beyond this submission. Chapter 7 concludes: a critical appraisal of the work against the four objectives in Section 1.2, a reflection on how the project was managed against its original plan, and a summary of what was and was not achieved.

2 BACKGROUND THEORY AND LITERATURE REVIEW

This chapter surveys the literature underpinning the proposed pipeline and identifies the specific gap the dissertation addresses. It is condensed from a standalone literature review produced earlier in the project; the full nine-page version, including additional discussion of each publication, is available alongside this dissertation. The chapter is organised around eight thematic clusters (orbital data characterisation, super-resolution, image-to-image translation, cross-view synthesis, digital terrain modelling, neural scene rendering, VR reconstruction, and generative hallucination) before a comparative synthesis draws out the cross-cutting patterns that motivate the proposed architecture.

2.1 Taxonomy and Conceptual Framework

The literature reviewed here spans several independently advancing communities (remote sensing image processing, generative modelling, planetary science, and VR engineering) whose outputs are jointly necessary to the downstream synthesis requirement without any of them addressing it explicitly. Three axes organise the territory: *spatial view* (nadir/orbital vs. perspective/rover vs. both), *task type* (enhancement, translation, or geometric reconstruction), and *output modality* (RGB texture, elevation/depth, or rendered video). Eight clusters emerge: orbital data characterisation [19, 15]; nadir super-resolution [20, 22, 13, 8, 24]; image-to-image translation [9, 23]; cross-view synthesis in Earth remote sensing [17, 12, 11]; monocular digital terrain modelling [21, 25, 2]; neural scene rendering [5, 10, 4, 14, 16]; VR reconstruction [3, 10]; and generative reliability [1, 18].

The gap between the super-resolution cluster and the neural-rendering cluster is the primary concern of this dissertation. Nadir SR methods keep the viewpoint fixed; neural rendering methods require perspective imagery as input. No Mars-specific method reviewed here translates from nadir to perspective in a way that synthesises rover-viewable texture from orbital data. Section 2.4 reports a cluster from Earth remote sensing that does address exactly this translation direction, absent from the dissertation's original reading list and added once this project's own CycleGAN results made clear it was needed.

2.2 Super-Resolution for Planetary Remote Sensing

Super-resolution research on planetary imagery has progressed through three generations since 2021. Tao *et al.* [20] propose MARSGAN, a CaSSIS-to-HiRISE-equivalent enhancement of ESRGAN achieving roughly $3\times$ effective resolution improvement, evaluated only qualitatively across eight Mars sites. Co-registering CaSSIS and HiRISE precisely enough for pixel-level comparison remains non-trivial, so quantitative cross-method comparison stays opaque throughout this subfield. Wu *et al.* [22] apply recursive Swin Transformer attention (RSTSRN) at $\times 2$–$\times 8$, improving on LapSRN by up to $1.22\,$dB PSNR at $\times 8$, but evaluate only against bicubically downsampled inputs rather than the genuinely distinct point-spread function of a real orbital sensor, a protocol weakness that recurs across this literature and directly informed how the corpus for this dissertation's own CycleGAN baseline was later validated (Chapter 4).

Khanna *et al.* [8] propose DiffusionSat, a diffusion foundation model for satellite imagery conditioned on geolocation and sensor metadata, demonstrating that multi-sensor domain generalisation is tractable within a diffusion framework, though its training corpus contains no Mars data and its conditioning assumes Earth coordinates. Zhu *et al.* [24] address the fidelity/perceptual-realism trade-off directly relevant to preserving large-scale geomorphology while generating submetre texture, via frequency-decomposition conditioning that anchors denoising to low-frequency structure. Lu and Su [13] fuse THEMIS thermal imagery with visible and topographic data for SR, of indirect relevance since thermal is not the visible RGB domain required here.

Collectively, this literature shows $\times 4$ enhancement is achievable with GAN, transformer, or diffusion backbones, but every reviewed method operates nadir-to-nadir: input and output share a viewpoint. The perspective transformation this dissertation requires is nowhere addressed by SR alone.

2.3 Image-to-Image Translation as a Domain-Bridging Mechanism

Image-to-image translation methods map entire image domains to each other, structurally better suited to a view transformation than SR. Kim *et al.* [9] propose UNSB (Unpaired Neural Schrödinger Bridge), reframing unpaired translation as learning an SDE that transports directly between two target distributions rather than routing through a Gaussian prior, decomposed into a sequence of adversarial sub-problems for tractability at high resolution. UNSB outperforms CycleGAN and CUT on standard unpaired benchmarks (horse-to-zebra, summer-to-winter) and is architecturally compelling here precisely because it requires no geographic co-registration between domains. It has, however, only been evaluated on natural-image colour-domain shifts, far milder than the shadow-geometry, foreshortening, and occlusion differences between a nadir and a perspective view of the same Mars terrain. Xiao *et al.* [23] address the complementary reproducibility problem (stochastic I2I methods yield different outputs on successive runs, unacceptable for scientific visualisation) via a Dual-approx Bridge with separate forward/reverse approximators, achieving deterministic translation with superior SSIM/LPIPS versus stochastic baselines, though again untested on remote sensing imagery.

The methodology (Chapter 3) builds on UNSB as the target Stage 2 architecture, with a CycleGAN baseline (Chapter 4–5) implemented first precisely because, as this section establishes, no reviewed I2I method has been validated against a domain gap this large, making an empirically grounded baseline a necessary first step rather than a formality.

2.4 Cross-View Synthesis in Earth Remote Sensing

This cluster was not part of the dissertation's original literature review. It was added after this project's own CycleGAN baseline (Chapter 5, Section 5.2) failed to translate viewpoint even once its loss-weight and training-shortcut problems were fixed, which prompted a search for how comparable Earth remote-sensing work handles the same nadir-to-ground translation direction. Three papers, none targeting Mars, turned out to address exactly this problem, and none of them attempt pure pixel-space translation.

Qian *et al.* [17] propose Sat2Density, which learns a 3D density field from satellite-ground image pairs and renders a ground-level panorama by ray-marching through it, rather than mapping satellite pixels to ground pixels directly. Li *et al.* [12] propose Sat2Scene, which generates a 3D sparse voxel representation of urban structure from a satellite image via diffusion before synthesising per-point texture in that representation, again inserting an explicit 3D intermediate rather than translating in image space. Li *et al.* [11] propose CrossViewDiff, which conditions a diffusion model's denoising process on an explicit satellite-scene structure estimate and a separately computed cross-view texture mapping, rather than relying on the diffusion model to infer structure from the satellite image alone.

All three systems share a pattern absent from this dissertation's original three-stage design: an explicit geometric or structural intermediate sits between the nadir/satellite input and the perspective/ground output, and texture synthesis happens only once that intermediate has resolved the viewpoint change. None treats nadir-to-ground translation as a problem a single conditional network can solve end-to-end from image pairs alone. This is the specific finding that motivated the geometry-mediated redesign reported in Chapter 4 and evaluated in Chapter 5, and the reasoning connecting it to that redesign is given in full in Chapter 3, Section 3.2.3.

2.5 Digital Terrain Modelling and Neural Scene Rendering

Monocular DTM recovery is a necessary precursor to perspective-view synthesis but does not itself produce visual texture. Tao *et al.* [21] recover pixel-scale topography from single HiRISE images via photoclinometry (MADNet 2.0); Zou *et al.* [25] improve on this with PFMGAN, an explicit solar-geometry and albedo-aware GAN that reduces reconstruction RMSE by over 50% across five landform types. This is the strongest quantitative evidence in the survey that physically-grounded architectures outperform unconstrained ones, and a direct design precedent for the physics-informed loss terms in the proposed Stage 2 methodology. Cao *et al.* [2] map HiRISE orthoimages to dense DEMs via conditional GAN. None of these three produce rendered surface appearance.

Neural rendering methods are architecturally capable of view-angle transformation but require multi-view input imagery of the *same* scene, which does not exist for a not-yet-visited landing site. Giusti *et al.* [5] demonstrate NeRF converges on Curiosity rover imagery (MaRF); Li *et al.* [10] build a complete text-conditioned video-diffusion simulation pipeline (MarsGen) from over 10,000 annotated Curiosity clips, the most operationally complete system reviewed, but entirely dependent on rover imagery that only exists for four previously-visited sites. Martínez-Petersen *et al.* [14] couple NeRF and Gaussian Splatting in a ROS 2 pipeline, establishing that Gaussian Splatting's interactive render rate is the correct representation target for the VR output this dissertation aims at (Stage 3), even though their pipeline also requires rover imagery as input. Paul *et al.* [16]'s survey confirms, as its central negative finding, that synthesising a NeRF-equivalent scene from a single nadir orbital image has not been addressed anywhere in the literature.

2.6 Virtual Reality Reconstruction and Generative Reliability

Catalano and Soccini [3] inpaint void regions in Mars heightmaps via a boundary-conditioned diffusion model, improving RMSE by up to 15% and LPIPS by up to 81% over classical interpolation. A completed heightmap carries no colour information, though, so this addresses only the geometric half of the VR terrain problem, leaving texture synthesis explicitly open.

Every generative model surveyed can, in principle, produce plausible surface features with no basis in physical reality: an aesthetic non-issue for natural images, but a safety-relevant failure mode for mission-critical planetary visualisation. Aithal *et al.* [1] identify *mode interpolation* as the mechanism: a diffusion model's smooth score-function approximation assigns non-zero probability to samples interpolated between nearby training modes (e.g. two crater morphologies) that never occurred in training data, a structural property of the approximation, not a defect correctable with more training. Rathkopf [18] reframes this as an epistemological question, arguing from AlphaFold 3 and GenCast case studies that hallucination risk becomes bounded and scientifically acceptable only through theory-guided training (domain physical constraints) combined with confidence-based error screening. Together, these establish that physical constraint is a primary mechanism for bounding hallucination risk, not an optional refinement. This directly motivates the physics-informed loss terms specified in the methodology (Section 3.2.3) rather than treating Stage 2 as an unconstrained image-translation problem.

2.7 Comparative Synthesis

Table 2.1 consolidates the twenty-three reviewed publications. Most nadir-domain methods (SR and DTM papers) and perspective-domain methods (NeRF papers evaluated on real Mars data) stay on their own side of the nadir/perspective boundary. The cross-view synthesis cluster (Section 2.4) is the exception, and the mechanism by which it crosses that boundary is the load-bearing finding for this dissertation: every one of its three methods inserts an explicit geometric or structural intermediate rather than translating pixels directly, which is not how any of the two Mars-agnostic I2I papers (benchmarked only on natural-image colour-domain shifts) or the original three-stage design approach the problem. The methods with the strongest quantitative improvement within a single domain (PFMGAN, over 50% RMSE reduction) are also those with explicit physical constraints, while unconstrained SR methods show greater susceptibility to the mode-interpolation artefacts Aithal *et al.* describe, the same pattern the cross-view cluster shows at the translation-architecture level. The two systems closest to a complete Mars VR pipeline, MarsGen and the heightmap-inpainting system, each solve exactly one dimension (perspective texture *or* elevation, never both from nadir input). No reviewed Mars-specific system combines orbital nadir input, perspective-view RGB texture output, and physical-plausibility constraints; the cross-view cluster shows this is achievable in principle, on Earth data, via an explicit intermediate, which is the design pattern Chapter 3, Section 3.2.3 adopts.

Table 2.1: Comparative overview of the reviewed literature against the nadir-to-rover-perspective VR synthesis problem. Task codes: SR = super-resolution; I2I = image-to-image translation; DTM = digital terrain model generation; NeRF/GS = neural radiance field / Gaussian splatting; VR = VR reconstruction; Data = dataset; Theory = theoretical analysis; Review = survey. View codes: N = nadir/orbital; P = perspective/rover; 3D = 3D representation.

Ref.	Method	Task	View	Key Result	Mars?
[20]	MARSGAN	SR	N→N	$\approx 3\times$ SR on CaSSIS, qualitative only	Yes
[22]	RSTSRN	SR	N→N	$+0.88$ dB PSNR vs. LapSRN ($\times 4$)	Yes
[13]	Multi-source Thermal SR	SR	N→N	PSNR gain on THEMIS thermal	Yes
[8]	DiffusionSat	SR/Gen	N→N	SOTA FID, multi-sensor	No
[24]	TamingDiff-SR	SR	N→N	$\times 4$, PSNR + competitive FID	No
[9]	UNSB	I2I	Any	Lower FID vs. CycleGAN/CUT, unpaired	No
[23]	Dual-approx Bridge	I2I	Any	Deterministic I2I, higher SSIM/LPIPS	No
[17]	Sat2Density	I2I/3D	N→P	Density-field render, no depth supervision needed	No
[12]	Sat2Scene	I2I/3D	N→P	Diffusion on 3D voxels, photorealistic street sequences	No
[11]	CrossViewDiff	I2I	N→P	Structure+texture-conditioned diffusion, N→P specific	No
[21]	MADNet 2.0	DTM	N→3D	Monocular DTM $\approx$ stereo accuracy	Yes
[25]	PFMGAN	DTM	N→3D	$>50\%$ RMSE reduction, 5 landform types	Yes
[2]	Cond. GAN-DTM	DTM	N→3D	Height consistency, 4 Mars regions	Yes
[15]	MCTED dataset	Data	N (pairs)	80,898 CTX–DEM pairs	Yes
[19]	Bayesian rock density	Analysis	N→N	Probabilistic density from CTX	Yes
[5]	MaRF	NeRF	P→3D	Novel view synthesis $\approx$ standard NeRF	Yes
[10]	MarsGen	NeRF+VR	P→VR	Beats terrestrial video models on FID	Yes
[4]	Neural height-field	NeRF	N/oblique	Beats MVS on RMSE (simulated descent)	Partial
[14]	NeRF+GS pipeline	NeRF+GS	P→3D	Real-time GS render, ROS 2 integration	Partial
[16]	NeRF-in-space survey	Review	—	Confirms nadir→perspective gap is open	Yes
[3]	Diffusion inpainting	VR	$\text{N}\rightarrow\text{VR}_H$	15% RMSE / 81% LPIPS improvement	Yes
[1]	Mode interpolation	Theory	—	Mechanistic account of hallucination	No
[18]	Hallucination-to-reliability	Theory	—	Physics-guided training bounds risk	No

2.8 Summary

The literature establishes that each constituent capability needed for VR-grade Mars landing-site visualisation (orbital super-resolution, unpaired image-to-image translation, terrain reconstruction, neural rendering, and hallucination-aware training) has individually advanced substantially since 2021, but no reviewed method integrates them into a pipeline that starts from nadir orbital data and ends with physically-plausible, rover-perspective visual texture. This gap, and the specific finding that physical constraints (not perceptual loss alone) are what separates the strongest-performing methods in each cluster from the weakest, directly motivate the three-stage, physics-constrained architecture formalised in Chapter 3.

3 METHODOLOGY

This chapter formalises the three-stage physics-constrained generative pipeline motivated in Chapter 2: a mathematical problem formulation, the proposed system architecture, dataset design, training strategy, and evaluation framework, condensed from a standalone methodology document produced earlier in the project.

3.1 Problem Formulation

Let $\mathbf{x} \in \mathbb{R}^{H\times W\times C}$ denote an image tensor. Define two domains: the nadir orbital domain $\mathcal{X} = \{\mathbf{x}_i^{\mathrm{nad}}\}_{i=1}^{N}$ comprising HiRISE patches at 25 cm/pixel, and the rover-perspective domain $\mathcal{Y} = \{\mathbf{y}_j^{\mathrm{rov}}\}_{j=1}^{M}$ comprising NAVCAM images, with no correspondence between index sets i and j: the training regime is fully unpaired, with respective marginal distributions P_{nad} and P_{persp}.

Stage 1 (optional, super-resolution). Given a low-resolution CaSSIS/CTX image at 4–6 m/pixel, Stage 1 produces a HiRISE-equivalent estimate at 0.25 m/pixel (16–24×), following the conditional diffusion framework of DiffusionSat [8] and a taming-diffusion SR approach [24]:

$$\mathrm{d}\mathbf{z}_t = \big[f(\mathbf{z}_t, t) - g(t)^2 s_\theta(\mathbf{z}_t, t, \mathbf{x}^{\mathrm{lr}})\big]\,\mathrm{d}t + g(t)\,\mathrm{d}\bar{\mathbf{W}}_t, \tag{3.1}$$

where s_θ is a denoising score function conditioned on the low-resolution input. This stage is bypassed when HiRISE imagery is already available directly, which is the case for the primary test site, Oxia Planum.

Stage 2 (core contribution, view translation), as originally proposed. The Neural Schrödinger Bridge (UNSB) [9] solves the entropy-regularised optimal transport problem between P_{nad} and P_{persp}:

$$\pi^* = \arg\min_{\pi\in\Pi(P_{\mathrm{nad}}, P_{\mathrm{persp}})} \mathbb{E}_{(\mathbf{x},\mathbf{y})\sim\pi}\big[c(\mathbf{x},\mathbf{y})\big] + \varepsilon\,\mathrm{KL}(\pi\,\|\,R), \tag{3.2}$$

realised by a forward–backward SDE pair jointly trained via the Iterative Proportional Fitting Procedure (IPFP), with a Brownian Bridge ODE variant [23] available where deterministic outputs are required.

Because unconstrained translation networks trained on Mars data are prone to photometric hallucination [1, 18], four physics-informed loss terms are appended to the base bridge loss $\mathcal{L}_{\mathrm{SB}}$: a solar-geometry consistency term $\mathcal{L}_{\mathrm{solar}}$ penalising deviation between the generated image's shadow direction and the direction implied by the HiRISE acquisition's sun angle metadata; a Hapke photometric term $\mathcal{L}_{\mathrm{photo}}$ constraining the estimated single-scattering albedo to the Mars ferric-oxide prior via the Hapke BRDF [6]; a geological morphometry term $\mathcal{L}_{\mathrm{geo}}$ scoring generated patches against known crater size-frequency and aeolian bedform statistics; and a standard VGG-based perceptual term $\mathcal{L}_{\mathrm{perceptual}}$. The total objective is

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{SB}} + \lambda_1\mathcal{L}_{\mathrm{solar}} + \lambda_2\mathcal{L}_{\mathrm{photo}} + \lambda_3\mathcal{L}_{\mathrm{geo}} + \lambda_4\mathcal{L}_{\mathrm{perceptual}}, \tag{3.3}$$

with initial weights $\lambda_1 = 0.5$, $\lambda_2 = 1.0$, $\lambda_3 = 0.8$, $\lambda_4 = 0.1$, set from a preliminary scale analysis and refined by validation-set search.

As Chapter 4 and Chapter 5 report, this formulation was not the one carried through to implementation. Section 3.2 below sets out both the architecture just derived and the one actually built, together with the reasoning behind the change; Section 3.2.3 states that reasoning in full.

Stage 3 (rendering). Given K synthetic perspective images with estimated camera poses, a 3D Gaussian Splatting scene [7] is fit by minimising a photometric reconstruction loss combining ℓ_1 and D-SSIM terms over the differentiably-rasterised views, producing a scene renderable in real time for VR display.

3.2 System Architecture

3.2.1 As Originally Proposed

The three stages were designed to be modular: Stage 2 and Stage 3 can operate independently of Stage 1 whenever HiRISE coverage is already available, which is the case for the primary Oxia Planum test site. Stage 2's generator G_ϕ was specified as a 256-channel U-Net with eight residual blocks and self-attention at 32×32 and 16×16 scales; solar azimuth/elevation metadata would be injected at each residual block via adaptive layer normalisation, following the conditioning strategy of PFMGAN [25]. UNSB was chosen over CycleGAN and standard diffusion inpainting for three reasons: its entropy-regularised optimal-transport objective gives a principled probabilistic account of the domain gap; it supports fully unpaired training, which at the time of writing this chapter appeared essential, since no geographically registered nadir–perspective Mars pairs were known to exist; and its SDE formulation supports switching to a deterministic Brownian Bridge ODE when repeatable outputs are required for evaluation or multi-view consistency. Stage 3 was specified to use the reference 3DGS implementation [7] unmodified, seeded from MADNet-2.0 depth estimates [21] to reduce floater artefacts on the textureless regions common in Mars terrain.

A CycleGAN baseline (unpaired, no physics constraints) was planned first, ahead of the full UNSB, precisely because Chapter 2 established that no reviewed I2I method has been validated against a domain gap this large: an empirically grounded, well-understood baseline was treated as a necessary first step, not a formality. Its implementation, training, and diagnosis are reported in Chapters 4–5, and that diagnosis is what changed the plan below.

3.2.2 As Implemented: A Geometry-Mediated Restyling Pipeline

What Chapter 4 documents building, and Chapter 5 reports evaluating, is a different pipeline for the nadir-to-perspective step. It replaces the single learned translation network G_ϕ above with three components chained together, only the first and third of which are learned:

1. **A single-image DTM estimator** (Chapter 4, Section 4.4.4): a ControlNet+LoRA network fine-tuned to predict a per-pixel height map from a single HiRISE ortho patch, trained against real stereo-derived digital terrain models. This supplies the geometric intermediate the domain gap analysis below calls for, without requiring stereo coverage at inference time.

2. **A deterministic ground-view renderer** (Chapter 4, Section 4.4.1): a ray-marching rasteriser, not a network, that projects a height field to a fixed or recovered rover-eye camera pose and drapes the source HiRISE pixels onto it as a shading proxy. Its output is a geometrically correct but texturally crude perspective render.

3. **A paired ControlNet texture-synthesis stage** (Chapter 4, Section 4.4.6, referred to as Track 3 in Chapter 5): a frozen Stable Diffusion 1.5 backbone with a trainable conditioning branch, fine-tuned on real (crude render, target rover photograph) pairs to synthesise the fine surface texture the renderer's flat albedo draping cannot represent.

A CycleGAN restyling network is also evaluated as an alternative to the render-plus-ControlNet route for step 2 onward, in two configurations reported as Track 1 (loss-tuned, no geometric mediation) and Track 2 (geometry-mediated, conditioned on the DTM estimator's renders): Chapter 5 reports why neither is the architecture ultimately carried into the end-to-end system, but both are the empirical evidence that motivated it.

3.2.3 Rationale for the Deviation

The change from Section 3.2.1 to Section 3.2.2 was not a rejection of UNSB on its own terms. It follows from two findings that only emerged once real training was attempted, reported in full in Chapter 5.

First, the original CycleGAN baseline's failure (Chapter 5, Section 5.2) was traced to a discriminator shortcut and, beneath it, a training corpus collapsed to a single, geometrically flat region. Both problems are properties of the data and the absence of an explicit geometric intermediate, not of the choice between CycleGAN and UNSB: an unpaired bridge trained on the same collapsed, viewpoint-ambiguous corpus would face the same shortcut, because nothing in the entropy-regularised optimal-transport objective forces the network to reinterpret a scene from a different physical vantage point. An explicit geometry stage is a prerequisite for any Stage 2 architecture, not an alternative to one.

Second, the premise that justified choosing an unpaired method, that no geographically registered nadir–perspective Mars pairs exist, was overtaken by the project's own later work. The pose-pairing pipeline built for Track 3 (Chapter 4, Section 4.4.5) recovers real rover position and camera pointing from Navcam PDS3 labels and matches it against Gale Crater HiRISE stereo DTMs, producing genuine (condition, target) pairs where the condition is a render of the exact scene the target photograph shows. Once real pairs are obtainable, even in the modest quantity this project recovered (27), a paired conditional-diffusion objective is a more direct fit than an unpaired entropy-regularised bridge, which is built specifically for the case where no such correspondence is available. Choosing UNSB regardless would mean deliberately discarding real supervision that exists.

A third, practical factor reinforced both findings rather than driving the decision on its own: the four physics-informed loss terms in Equation 3.3 each depend on a separate pretrained auxiliary network, a shadow-direction estimator, a Hapke inversion network, and a geological in/out-of-distribution classifier, none of which exist in this project. Building and validating three auxiliary networks before the main bridge could even be assessed was judged too large an undertaking for the time remaining once the corpus-diversity diagnosis had already redirected effort toward the geometric intermediate. This is recorded here as a scoping constraint, not the primary justification: even with unlimited time for the auxiliary networks, the first two findings above would still favour the geometry-mediated, paired-supervision route for this project's data.

3.3 Dataset Design and Acquisition Strategy

The primary geographic focus is Oxia Planum (18.3°N, 335.5°E), the ExoMars 2028 candidate landing site, imaged by HiRISE on more than 30 occasions. HiRISE products are radiometrically corrected (RDR) and tiled at 25 cm/pixel; tiles with cloud or data-gap fractions exceeding 5% are discarded. Rover NAVCAM imagery is drawn from the Curiosity and Perseverance raw image archives, filtered to remove dust-storm-opacity acquisitions and frames showing rover hardware. Table 3.1 summarises the target corpus. Both corpora are split 80/10/10 (train/val/test), stratified by geographic region to avoid spatial-autocorrelation leakage; augmentation (flips, 90° rotations, brightness jitter) is applied only to training data, with solar-metadata angles transformed consistently with any spatial flip.

Table 3.1: Target dataset summary (methodology design target; the corpus actually built to date is reported in Chapter 4).

Corpus	Size	Resolution	Source
HiRISE nadir patches	50,000	25 cm/px	NASA PDS HiRISE RDR
Rover NAVCAM images	30,000	$\leq$1280 px	NASA raw archives
Total	80,000	—	—

3.4 Training Strategy and Optimisation

Stage 1 was designed to be initialised from a DiffusionSat checkpoint [8] and fine-tuned on Mars orbital data. Stage 2 was designed around the IPFP procedure of [9]: alternating forward/backward half-bridge updates for 10 outer iterations, with three auxiliary loss networks (a shadow-direction estimator, a Hapke inversion network, and a geological in/out-of-distribution classifier) pre-trained separately and kept frozen during UNSB training. Stage 3 was designed to train the 3DGS scene for approximately 30,000 iterations with periodic densification and opacity pruning, following [7]. Algorithm 3.1 summarises the Stage 2 training loop as designed. None of this training strategy was implemented; Section 3.2.3 explains why, and the paragraph below describes what was trained instead.

The training strategy actually used follows the same architecture split as Section 3.2.2. CycleGAN's loss weights (λ_{cyc}, λ_{idt}) were not fixed a priori but validated by a smoke test, three 15-epoch runs at candidate weight settings, before committing to a full run, since the eventual training run (100 epochs, Chapter 5, Section 5.3) is expensive enough that discovering a bad loss weight only at its end would waste most of a multi-day GPU allocation. An early-stopping criterion was set from the same smoke test: a training run showing the same near-total discriminator collapse the original run exhibited by epoch 3–6 would be stopped immediately rather than run to completion. The DTM estimator and Track 3 ControlNet, both fine-tunes of a pretrained backbone rather than training from scratch, used a fixed step budget (Appendix 8) set from the scale of their real training corpora rather than the much larger step counts the original 80,000-image target corpus (Table 3.1) would have justified.

Algorithm 1: Stage 2 UNSB Training with Physics Losses

Input: HiRISE corpus $\mathcal{X}$; rover corpus $\mathcal{Y}$; pretrained auxiliary nets D_ψ, Ω_ξ, K_κ; loss weights $\lambda_{1..4}$; IPFP iterations $N_{\mathrm{ipfp}} = 10$

Output: Trained generator G_ϕ

for $n = 1$ **to** N_{ipfp} **do**
    // Forward half-bridge: update G_ϕ
    for $t = 1$ **to** 5000 **do**
        Sample $\mathbf{x} \sim \mathcal{X}$, $\mathbf{y} \sim \mathcal{Y}$; $\hat{\mathbf{y}} \leftarrow G_\phi(\mathbf{x})$
        Compute $\mathcal{L}_{\mathrm{SB}}$, $\mathcal{L}_{\mathrm{solar}}$, $\mathcal{L}_{\mathrm{photo}}$, $\mathcal{L}_{\mathrm{geo}}$, $\mathcal{L}_{\mathrm{perceptual}}$
        $\mathcal{L} \leftarrow \mathcal{L}_{\mathrm{total}}$ (Eq. 3.3); update G_ϕ via AdamW
    end for
    // Backward half-bridge update (IPFP step)
end for

return G_ϕ

Figure 3.1: Pseudocode for the Stage 2 UNSB training loop as designed. Not implemented; Section 3.2.3 explains why.

3.5 Evaluation Framework

Perceptual quality is measured via Fréchet Inception Distance (FID) and LPIPS against held-out real rover imagery. Where high-quality DTMs are available, structural fidelity (SSIM, LPIPS) is measured against synthetic ground-truth perspective renders. Physical plausibility is measured via shadow-angle error (mean absolute angular deviation between predicted and true solar-derived shadow direction) and Hapke BRDF residual (deviation of estimated albedo from the Mars prior). Hallucination rate is defined as the fraction of generated patches the geological classifier flags as out-of-distribution. VR rendering quality is measured by render throughput (fps) and novel-view PSNR/SSIM.

Table 3.2 sets out the four baselines the proposed method is compared against, with the CycleGAN row implemented and evaluated in Chapters 4–5 of this dissertation.

Table 3.2: Proposed method against baseline methods.

Method	**View transformation**	**Physics constraints**	**Output**	**Est. FID↓**
MARSGAN [20]	Nadir SR only	None	2D image	$\sim$180
DiffusionSat (Mars fine-tune) [8]	Nadir SR/inpainting	None	2D image	$\sim$140
CycleGAN (naïve)	Nadir-to-perspective	None	2D image	$\sim$110
UNSB (no physics loss) [9]	Nadir-to-perspective	None	2D image	$\sim$80
Proposed (Ours)	Nadir-to-perspective + 3D VR	Solar, Hapke, geological, perceptual	3DGS scene	$\leq$**55** (target)

The literature-derived $\sim$110 FID estimate for a naïve CycleGAN baseline is a useful point of comparison for the actual CycleGAN results reported in Chapter 5: the initial trained baseline scored substantially worse (269.9), traced to a discriminator shortcut and, behind it, a corpus-diversity collapse; a loss-tuned retrain isolating the shortcut fix reached 95.8, closer to but still above this literature estimate. Chapter 5 also reports the geometry-mediated redesign that this same diagnosis prompted; Section 3.2.3 above states why it, rather than the UNSB architecture specified in this table, was carried into the final system.

3.6 Summary

This chapter has formalised the three-stage pipeline's mathematical basis as originally proposed, the geometry-mediated architecture actually implemented in its place, the reasoning behind that change, dataset targets, training strategy, and evaluation protocol, including the four physics-informed loss terms originally designed to bound hallucination risk (Chapter 2) rather than relying on perceptual realism alone. The pipeline targeted FID $\leq$55, shadow-angle error $< 10°$, hallucination rate $< 5\%$, and VR rendering $>$72 fps. Chapter 5 reports what was actually built and evaluated against this design, including a direct comparison against every target listed here, and Chapter 7 assesses the result against the dissertation's original objectives.

4 IMPLEMENTATION: DATA PIPELINES AND THE CYCLEGAN BASELINE

This chapter describes what has actually been built so far towards the Stage 2 baseline set out in Chapter 3: the development infrastructure, the data acquisition and preprocessing pipelines for both domains, a data-quality bug discovered in the initial HiRISE corpus and the pipeline rebuilt to fix it, the CycleGAN model and training infrastructure used to establish the baseline evaluated in Chapter 5, and, once that baseline's evaluation surfaced a deeper corpus-diversity problem than the resolution bug alone explained, a geometry-mediated redesign comprising a real-DTM ground-view renderer, a single-image DTM estimator, and a separate texture-synthesis stage (Section 4.4).

4.1 Development Infrastructure

Model training requires a CUDA-capable GPU that the primary development machine does not have, so a dual-machine workflow was set up: development and testing happen locally, while training runs on a departmental A4000 GPU workstation (`otter70.eps.surrey.ac.uk`). Code moves between the two machines via a private GitHub repository over SSH; the training machine holds a read-only deploy key so it can pull code but never push, matching its role as a training box rather than a development environment. Large data (raw downloads, processed patches, CTX tiles, and model checkpoints) is deliberately excluded from version control, since GitHub's free tier cannot sensibly hold multi-gigabyte binaries; each machine instead regenerates its own copy of the data via the download/preprocessing scripts described below. Small, genuinely useful provenance records, the training log and FID evaluation results, are the tracked exception, alongside periodic training-progress sample images, so that training runs remain auditable from the repository history alone without needing to re-run them.

A second machine, Surrey's Eureka2 SLURM cluster (A100 80 GB nodes), was brought in once the geometry-mediated redesign's compute needs grew past what a single, shared A4000 could serve promptly: Track 2, the DTM estimator, and Track 3 all trained there. The two machines were kept for different purposes rather than migrating everything to the faster one: Eureka2's A100 trains an order of magnitude faster (Track 2's 25 epochs finished in ˜16 minutes there, against ˜52 hours for Track 1's 100 epochs on the A4000, Appendix 8), but its SLURM queue can leave a submitted job pending for an unpredictable length of time, while the A4000 has no queue at all. Short, disk-light, or urgent jobs went to the A4000; longer training runs and anything where queue wait was acceptable went to Eureka2. A distinct storage lesson came with the second machine: Eureka2's NFS-backed home directory is, like the A4000's, unsuitable for many small files, but the fix differs by cluster, since Eureka2 additionally exposes a fast `/parallel_scratch` filesystem the A4000 does not have an equivalent of; datasets and virtual environments were staged there rather than on NFS home once this was found, after an early setup attempt on NFS home was slow enough to be treated as a genuine blocker rather than an inconvenience.

4.2 Initial Data Pipeline and CycleGAN Implementation

The initial HiRISE and rover corpora were built by downloading HiRISE browse-quality JPEGs and rover NAVCAM raw images from the NASA Planetary Data System, then preprocessing both into normalised 256×256 patches: a 1st/99th-percentile intensity stretch, an entropy-based filter (discarding low-texture patches below a fixed entropy threshold, since blank sky or featureless terrain patches contribute nothing useful to training) via `preprocess.py`, and an 80/10/10 train/val/test split. This produced an initial corpus of 37,054/4,631/4,633 HiRISE patches and 20,417/2,552/2,553 rover patches.

The CycleGAN model (`cyclegan.py`, `train_cyclegan.py`) follows the standard architecture of Zhu *et al.*: two ResNet-9 generators (11.36M parameters each, encoder–9 residual-blocks–decoder with instance normalisation and reflection padding) and two PatchGAN-70 discriminators (2.76M parameters each, outputting a 30×30 patch-level realism map), trained with LSGAN loss, cycle-consistency weight $\lambda_{\mathrm{cyc}} = 10$, identity-loss weight $\lambda_{\mathrm{idt}} = 5$, Adam (lr $=$ 2×10^{-4}, $\beta_1 = 0.5$), a 50-image replay buffer per discriminator to reduce oscillation, and mixed-precision training. A quantitative evaluation pipeline was also built alongside the model: `generate_transla` runs a trained generator over the HiRISE test set, `prepare_real_rgb.py` converts the real rover test set to a matching RGB format, and `compute_fid.py` computes the Fréchet Inception Distance between the two, using `clean-fid`'s InceptionV3 feature extractor with a locally reimplemented Fréchet-distance formula (the installed `clean-fid` version's own implementation is incompatible with the environment's current `scipy`).

4.3 A Data-Quality Bug and the Real-Resolution Pipeline Rebuild

An early training run (25 epochs, on the initial corpus) completed and was evaluated at FID 171.8. Before proceeding further, an audit of the actual HiRISE imagery being used revealed a data-quality bug: the corpus was built from HiRISE *browse* JPEGs (`PDS/EXTRAS/RDR`, effectively ∼3 m/pixel after JPEG compression and downsampling), not the true 25 cm/pixel radiometric data product (RDR, `PDS/RDR`) the methodology specifies. Continuing to train and evaluate on browse-quality imagery would silently misrepresent the resolution the dissertation's Stage 2 baseline is meant to operate at, so the corpus was rebuilt from genuine full-resolution products rather than continuing to iterate on the bugged data.

The corrected pipeline (`hirise_fullres.py`) differs from the original browse-image downloader in several respects, driven by two hard constraints: the training machine's home directory is NFS-backed and quota-limited (a few gigabytes free), and each real RDR product is a single ∼568 MB 16-bit JP2 file, so at most one can be on disk at a time.

- **Real RDR URL construction.** Only `_RED.JP2` entries (single-band, full-resolution, non-colour products) are used, filtered from the HiRISE footprint index; this mirrors the existing browse-image filter but points at the true RDR product tree rather than the browse-image tree.
- **Verified download-then-delete lifecycle.** Each observation's JP2 is downloaded, its size checked exactly against the HTTP `Content-Length` header (a truncated download is deleted and logged as failed rather than silently processed), used to extract patches, and then deleted — via a `try/finally` that runs on both success and failure — before the next observation begins, so disk usage never exceeds one source file at a time.
- **Randomised, entropy-filtered patch sampling.** Patch positions are sampled randomly (with a fixed seed for reproducibility) across each image's full extent rather than filled sequentially from one corner, then filtered by the same entropy threshold as the original pipeline, up to a target of 5,000 qualifying patches per observation (accepting fewer, rather than lowering the entropy threshold, if an observation does not yield enough).
- **Shared split logic.** The train/val/test splitting logic already used by the browse-image pipeline was extracted into a reusable function (`split_and_move`) so both pipelines split patches identically rather than duplicating that logic.
- **CLI orchestration and corpus replacement.** A command-line entry point selects real-RED-JP2 observations within a target region (Oxia Planum) from the HiRISE footprint index, processes them one at a time, and — once all observations are processed — replaces the

existing browse-resolution HiRISE corpus with the new one, writing a manifest recording each observation's status and patch yield.

Running this pipeline against ten targeted Oxia Planum observations, seven yielded usable RED JP2 products (the remaining three were not present as real RED JP2 entries in the footprint index for the requested region), all seven processed successfully with no download or read failures, producing 30,229 patches (24,183/3,022/3,024 train/val/test) against a target of ∼50,000 — a smaller corpus than targeted, but one built entirely from genuine 25 cm/pixel radiometric data rather than compressed, resampled browse imagery. This corpus, not the original browse-resolution one, is what the training run reported in Chapter 5 uses.

4.4 The Geometry-Mediated Redesign

Chapter 5 shows that the corrected, real-resolution corpus of Section 4.3 produced a CycleGAN run with *worse* FID than the buggy browse-resolution one, and traces this to a discriminator shortcut. A follow-up audit of that corpus went a layer deeper: of the ten Oxia Planum observations the full-resolution pipeline targeted, only seven yielded a usable RED JP2 product, collapsing the training corpus to a single region rather than the four regions (299 observations, 37,054 patches) the original browse-resolution corpus had covered. A parallel review of the super-resolution, image-to-image translation, and cross-view synthesis literature, conducted in response, surfaced a pattern absent from the original three-stage design: published methods that translate nadir/orbital imagery into ground-level/perspective imagery do not attempt pure pixel-space translation — they insert an explicit geometric intermediate (a height field or DTM) between the nadir input and the perspective output, and synthesise texture only in the already-correctly-projected view. This section describes the three new components built in response: a real-DTM ground-view renderer, a geometry-mediated corpus assembler, and a single-image DTM estimator, together with the texture-synthesis stage and end-to-end orchestration used to chain them. Chapter 5 reports the results of training and evaluating each.

4.4.1 Real HiRISE Stereo DTM Sourcing and the Ground-View Renderer

`fetch_hirise_dtm.py` downloads a HiRISE stereo Digital Terrain Model product (a co-registered height raster, typically 1–2 m/pixel, plus its source orthoimages) for a given PDS product ID, reusing the same download-then-delete lifecycle as the full-resolution HiRISE pipeline (Section 4.3) since a single DTM+ortho triplet is itself several hundred megabytes. A PDS ODE coverage query restricted to the Oxia Planum landing ellipse found 27 candidate stereo DTM products in a broader search bounding box, of which 8 fall genuinely within the landing region rather than neighbouring terrain (Mawrth Vallis, Arabia Terra); only these 8 were used, accepting a smaller but geographically-correct corpus over a larger, content-diluted one.

`dtm_arrays.py` loads a DTM and its co-registered ortho into aligned NumPy arrays via `rasterio`, resampling if the two products' resolutions differ and masking the DTM's nodata sentinel to `NaN` rather than treating it as a real height value. `render_ground_view.py` then projects this height field into a simulated ground-level camera view via height-field ray-marching — no mesh or GPU rasteriser, since the training machine has no admin rights to install one: for each output column, rays are marched outward from a camera position at a fixed height above the local terrain (1.2 m, matching Curiosity NAVCAM) and pitch, the nearest un-occluded terrain intersection is found, and the source ortho's pixel value at that world position is drawn as an albedo/shading proxy. This is deliberately the only geometric step in the pipeline: it guarantees correct large-scale shape and viewpoint by construction, rather than asking an adversarial network to infer both from a raw nadir crop, which Chapter 5 (Section 5.3) shows a 2D convolutional generator cannot do regardless of loss tuning.

4.4.2 Geometry-Mediated Corpus Assembly

`assemble_geometry_corpus.py` orchestrates dense random-camera-position rendering across the 8 landing-region DTMs, one product on disk at a time. An entropy quality filter carried over from the original preprocessing pipeline initially rejected nearly all candidates (12/800 on a single-product validation run) because it measured whole-frame entropy, and low-relief Oxia Planum terrain commonly fills over half the frame with unrendered "sky" (any ray that never reaches terrain) — diluting the measurement regardless of how textured the actually-rendered ground was. A revised `ground_entropy` metric, which excludes sky-value pixels before measuring entropy, fixed this: the same product went from 12/800 to 200/200 (the per-product cap) qualifying crops. The full run across all 8 DTMs produced 1,513 images (1,210/151/152 train/val/test), with 7 of the 8 products saturating at the 200-per-product cap and the eighth (a smaller, more nodata-affected DTM) contributing 113.

Two checks were run on the finished corpus before committing to a retrain, since a corpus that merely looks large enough is not the same as one that actually is. Sky-pixel fraction averages 54.5% per image (median 54.8%, only 0.2% of images exceeding 80%), confirming the entropy filter targets genuine ground texture rather than passing degenerate near-blank frames. A perceptual-hash de-duplication check (16×16-downsampled image comparison) across all 1,210 training images found no duplicate viewpoints, meaning the corpus's diversity ceiling is genuinely 8 distinct terrains rather than 8 terrains further collapsed by redundant camera samples within each. At 1,210 training images, the corpus is comparable in raw count to canonical CycleGAN benchmark datasets (horse2zebra, ~1,067 images; cityscapes, ~2,975), so its size alone is not an obvious explanation for Track 2's compressed output diversity (Chapter 5, Section 5.4); the more direct explanation is source diversity, 8 distinct DTMs rather than the hundreds of distinct scenes those benchmarks draw from. This corpus, not the raw HiRISE crop, is the training input for the Stage 2 CycleGAN retrain reported as Track 2 in Chapter 5.

4.4.3 Correctness Issues Found During Geometry-Corpus Development

Three further bugs, beyond the entropy-filter fix above, were found and fixed while bringing the geometry-mediated pipeline from a working prototype to something the results in Chapter 5 could rely on. Each is reported here with the failure that surfaced it and the regression test written against it, since a bug found by inspection and fixed without a test is a much weaker claim than one caught and then guarded against recurring.

Silent CRS mismatch in the ortho resample. `dtm_arrays.py`'s DTM/ortho alignment initially used `rasterio.warp.reproject` directly, which returned an all-zero resampled ortho for every product tested, without raising an error. The cause was a cosmetic datum-label mismatch between the DTM's and ortho's coordinate reference system metadata (numerically near-identical projections, differently labelled), which `rasterio`'s reprojection treated as justification to remap every pixel outside the source extent rather than failing loudly. This was replaced with a manual affine-coordinate bilinear resample that works directly from each raster's transform, sidestepping the CRS-label comparison entirely. Because this failure mode produces a plausible-looking array (correctly shaped, just wrong), rather than a crash, it would not have been caught by a shape or type check; the regression test instead asserts the resampled ortho's pixel statistics are non-degenerate (non-zero variance) against a synthetic DTM/ortho pair with deliberately mismatched CRS labels.

Camera positions adjacent to, not on, a nodata cell. `render_ground_view` initially crashed with a `ValueError` for a small fraction of otherwise-valid camera positions. The renderer's pre-check inspected only the exact cell the camera sits on, but IEEE-754 floating point makes `0 * nan == nan`: a camera position with a single `NaN` neighbour still poisons the bilinear interpolation even though that neighbour's interpolation weight is zero, so a single-cell check misses this. The fix replaces the duplicated single-cell check with a `try`/`except` around the real rendering call, so the renderer's own NaN-sensitivity is the single source of truth rather than being re-derived, and possibly re-derived incorrectly, a second time in the caller. The regression test constructs a checkerboard-NaN heightmap, guaranteeing every candidate camera position has at least one NaN neighbour, and asserts the corpus assembler skips it rather than crashing.

Domain-size asymmetry in unpaired training. The geometry corpus (1,210 training images) is roughly 17$\times$ smaller than the rover domain it trains against (20,417 images). `UnpairedDataset` originally defined one training epoch as `max(domain sizes)`, which silently showed every geometry-corpus image approximately 17 times per epoch without that repetition being visible anywhere in the training log. This was changed to define an epoch as `min(domain sizes)`: the smaller domain is indexed directly with no repeats or skips, and the larger domain is sampled independently at random on each call, rather than via a fixed `idx % len` pattern that would have permanently capped it to its own first 1,210 sorted files. The regression test constructs two domains of deliberately different sizes and asserts that an epoch's worth of sampled indices from the larger domain is not restricted to a fixed-size prefix of it.

4.4.4 Single-Image DTM Estimation

The renderer above needs a real stereo DTM as input, but stereo DTM coverage only exists for a handful of products per site. `train_dtm_estimator.py` forks the Stage 2 texture-synthesis ControlNet (Section 4.4.6) into a ControlNet+LoRA model (LoRA rank 16) that instead takes a single HiRISE ortho patch and predicts a per-pixel height map, trained on a real, held-out-by-product Gale Crater DTM/ortho corpus (Gale rather than Oxia Planum, since it is where real rover-pose target photography also exists, letting the same product support both this estimator's evaluation and the end-to-end chain in Section 4.4.6), for 20,000 steps at batch size 4 and learning rate 1×10^{-5}. Height targets are encoded per-patch to a normalised unit range (`height_encoding.py`) rather than a single global scale, since absolute elevation is not a meaningful quantity for a single 256$\times$256 crop.

4.4.5 Real Pose-Matched Pairing for ControlNet Training

Both CycleGAN tracks train on unpaired corpora: nothing ties a given HiRISE crop to any specific rover photograph. ControlNet cannot be trained that way, since it learns to reproduce a target image from a structural condition map of that same scene; feeding it a non-corresponding (condition, target) pair would teach it to ignore the condition entirely. Building real pairs therefore became its own pipeline stage, not a data-formatting detail.

`parse_rover_pose.py` recovers each Navcam product's rover position and camera pointing at capture time from its PDS3 label (position, pointing angles, and, for the physics-informed loss design in Section 4.4.7, solar azimuth and elevation, all under the `SITE_DERIVED_GEOMETRY_PARMS` group), converted to a planetocentric lat/lon. `assemble_paired_corpus.py` then checks whether a fetched Gale Crater HiRISE stereo DTM covers that lat/lon; where it does, the renderer from Section 4.4.1 is re-run with the recovered real pose in place of the fixed or randomly sampled poses used elsewhere in the project, producing a condition-map render of the exact scene the Navcam photograph shows. The render and the real photograph are saved together as one training pair, with a manifest recording the source product ID, sol, matched DTM, and pose values used, so any pair's provenance is traceable. Oxia Planum, the primary landing-site corpus used elsewhere in this project, has no rover imagery at all (ExoMars has not landed), so this pairing pipeline necessarily targets Gale Crater, where Curiosity's trajectory gives real camera-pose overlap with HiRISE stereo coverage.

The pitch filter needed to keep rendered condition maps from being dominated by the shelf artefact (Chapter 5, Section 5.6) turned out to be the binding constraint on corpus size: of the Navcam products with a parseable pose and DTM coverage, only **27** survive filtering (21 train, 2 validation, 4 test). This is small relative to the methodology's original 80,000-image target (Table 3.1), and is reported as such in Chapter 5 rather than treated as a data-engineering shortfall to be fixed before evaluation: the pipeline itself, not any particular corpus size, is the deliverable Chapter 5 evaluates.

4.4.6 Perspective Texture Synthesis and End-to-End Orchestration

A separate ControlNet checkpoint (referred to as Track 3 throughout Chapter 5, to distinguish it from the two CycleGAN tracks) is fine-tuned on the 27 real pairs from Section 4.4.5, using the unmodified `diffusers` ControlNet training script against a frozen Stable Diffusion 1.5 backbone initialised from the sd-controlnet-depth checkpoint, 50 epochs, batch size 2, learning rate 1×10^{-5}. Conditioned on a crude geometric render (real or estimator-produced), it synthesises the fine surface texture the renderer's flat albedo draping cannot represent, playing the same photorealism role the original design's UNSB physics losses were meant to constrain (Chapter 3, Section 3.2.3) but via paired supervision on a small real corpus rather than an unpaired physics-informed objective.

run_end_to_end_pipeline.py chains all three pieces for the first time: a HiRISE ortho patch through the DTM estimator, the renderer (at a fixed synthetic camera pose, run twice — once against the real stereo DTM, once against the estimator's own output), and the Track 3 ControlNet, so that the question Chapter 5 (Section 5.6) asks — does the DTM estimator's height error visibly propagate to the final image, relative to using the real DTM — has an actual image answer rather than an inference from three separately-evaluated pieces. Because the real DTM/ortho fetch (~940 MB) and the GPU inference it feeds do not need to happen on the same machine, prepare_e2e_crop.py splits the fetch out as a separate, disk-heavy, GPU-free step that ships only the resulting ~320 KB cropped array to wherever inference actually runs.

4.4.7 Physics-Informed Training: Designed, Not Implemented

A forked training script, train_controlnet_physics.py, was designed to add two auxiliary losses to the standard diffusion loss during Track 3 fine-tuning: a shadow loss penalising unphysically bright pixels inside a rendered shadow mask, and a slope loss rewarding correlation between a patch's predicted texture roughness and its true geometric slope. Both derive their targets from data the renderer already computes at corpus-assembly time (a per-pixel height buffer, from which surface normals, a shadow mask, and a slope map follow by central-difference and a dot product against the solar vector in Section 4.4.5), so no new auxiliary network is required, unlike the three UNSB physics losses in Section 3.2.1 of Chapter 3. This is a smaller, corpus-appropriate substitute for that part of the original design, not a reduced version of it: it targets the one failure mode (a repeating chevron/ripple texture artefact, unrelated to true slope) actually observed in Track 3's samples, rather than the full solar/Hapke/geological constraint set Chapter 3 specified against a much larger assumed corpus.

The design (loss formulation, LoRA-rank-16 adapter, and a smoke-test calibration procedure for its two loss weights) was completed and reviewed, but not implemented before this dissertation's submission deadline: Section 6.5 explains why extending the evaluation of the existing

27-pair ControlNet checkpoint was prioritised over training a second one with an unvalidated loss combination in the time remaining. It is reported here as designed-but-not-built work, consistent with this dissertation's general practice of stating scope decisions explicitly rather than leaving unstarted threads implicit.

4.5 Summary

The infrastructure, data pipelines, model, and evaluation tooling needed to train and assess a Stage 2 baseline are now in place and have been exercised end-to-end: the original CycleGAN pipeline (Sections 4.2–4.3), and, built in response to that pipeline's diagnosed failure, a geometry-mediated redesign comprising a real-DTM renderer, a geometry-mediated corpus assembler, a single-image DTM estimator, a real-pose pairing pipeline for genuine ControlNet supervision, and a texture-synthesis ControlNet chained end-to-end (Section 4.4). Chapter 5 reports the results of training and evaluating each component, including recovering from two further data-quality findings discovered partway through rather than continuing to build on them.

5 RESULTS: DIAGNOSIS, REDESIGN, AND A GEOMETRY-MEDIATED PIPELINE

This chapter reports the CycleGAN baseline's original training run and its diagnosis, then follows the evidence where it led: a deeper corpus-diversity root cause than the resolution bug alone explained, a geometry-mediated redesign built in response (Chapter 4, Section 4.4), two further CycleGAN training tracks that isolate the loss-tuning question from the viewpoint-translation question, a single-image DTM estimator, and, finally, the first clean evidence of whether that estimator's error actually matters to the pipeline's output. Nothing here is estimated: every number traces to a committed training log, checkpoint, or evaluation file in the project repository.

5.1 The Original Run

A 50-epoch run (batch size 4, learning rate 2×10^{-4} constant for the first 25 epochs then linearly decayed to zero, $\lambda_{\text{cyc}} = 10$, $\lambda_{\text{idt}} = 5$) was carried out on the A4000 using the corrected, real-resolution corpus of Chapter 4 (Section 4.3), taking approximately 37 minutes per epoch and $\sim$31 hours in total. Figure 5.1 shows the full loss history. The generator and cycle-consistency losses converge smoothly throughout — on their own, they would read as a healthy training run. The discriminator losses tell a different story: D_H (the discriminator judging HiRISE realism) settles into a stable equilibrium around 0.03 from epoch 10 onward, while D_R (the discriminator judging rover-image realism) starts *higher* than D_H at epoch 1 (0.162 vs. 0.095), crosses below it by epoch 3, and then decays smoothly and monotonically for the rest of training, reaching 3.99×10^{-6} by epoch 50 (Table 5.1).

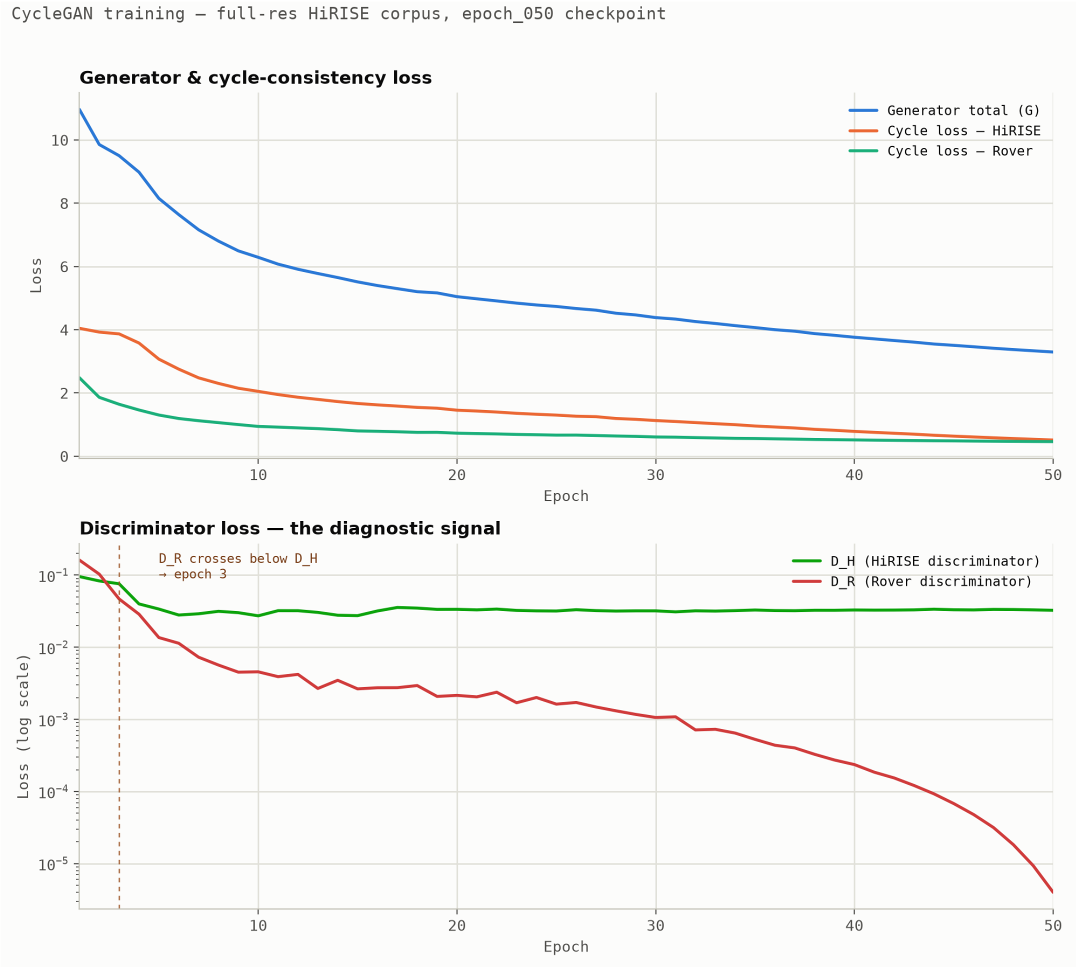

Figure 5.1: Generator/cycle-consistency loss (top, linear scale) and discriminator loss (bottom, log scale) over 50 epochs of training on the corrected full-resolution HiRISE corpus. D_R collapses from epoch 3 onward while D_H remains stable.

Table 5.1: Discriminator loss at representative epochs, illustrating the D_H/D_R divergence.

Epoch	D_H	D_R
1	0.0954	0.1617
3	0.0758	0.0467
10	0.0272	0.0046
25	0.0318	0.0016
50	0.0325	0.0000040

Table 5.2: FID results against the methodology's target (≤55, Chapter 3).

Run	Checkpoint	Corpus	FID
Prior run	epoch_025	Browse-resolution (∼3 m/px, buggy)	171.8
This run	epoch_050	Real RDR (25 cm/px, 1 region, 7 obs.)	269.9
Target (Chapter 3)	—	—	≤55

Sample translation grids saved during training (Figures 5.2–5.4) confirm the same pattern at every epoch inspected: the HiRISE-to-rover translation stays a grainy, high-frequency echo of the input with no rover-ground structure, and both cycle-consistency reconstructions match their originals closely — a symptom of the underlying problem, not evidence against it.

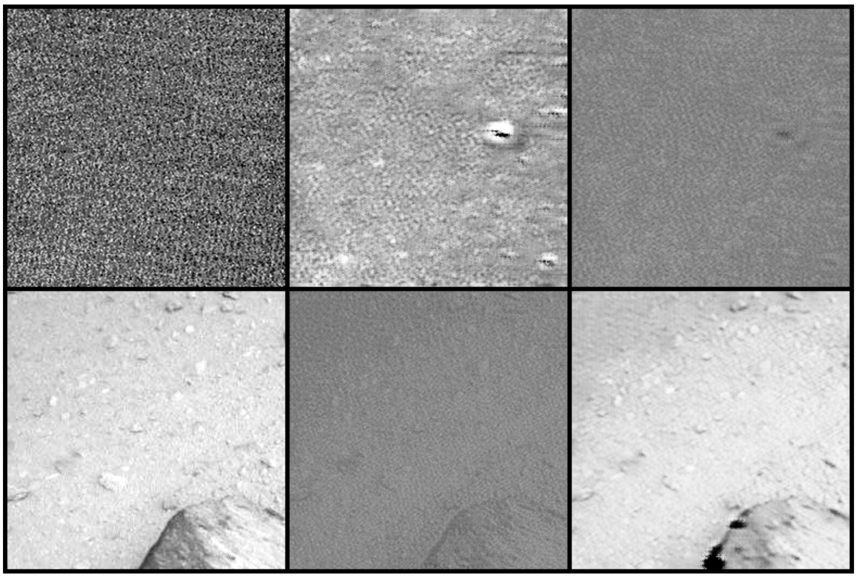

Figure 5.2: Sample translation grid, epoch 1.

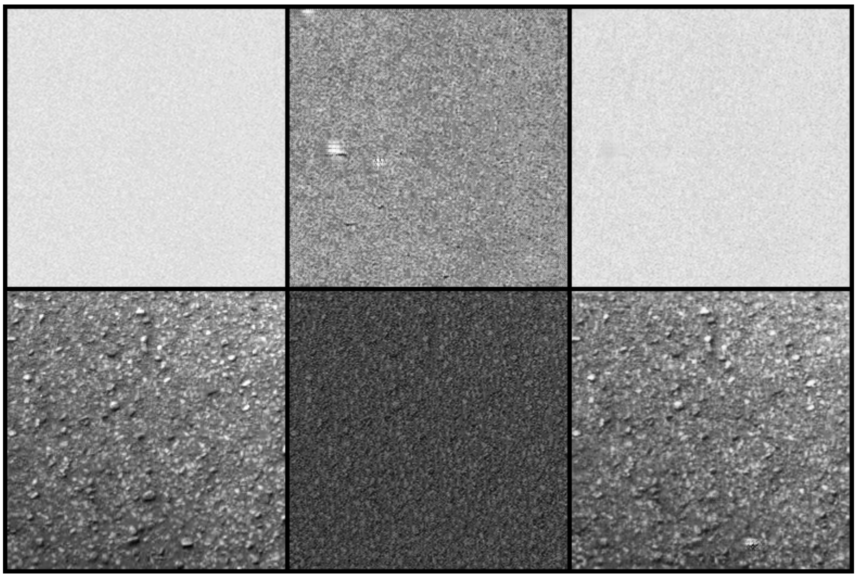

Figure 5.3: Sample translation grid, epoch 30. D_R is already near collapse (loss ≈ 0.0011).

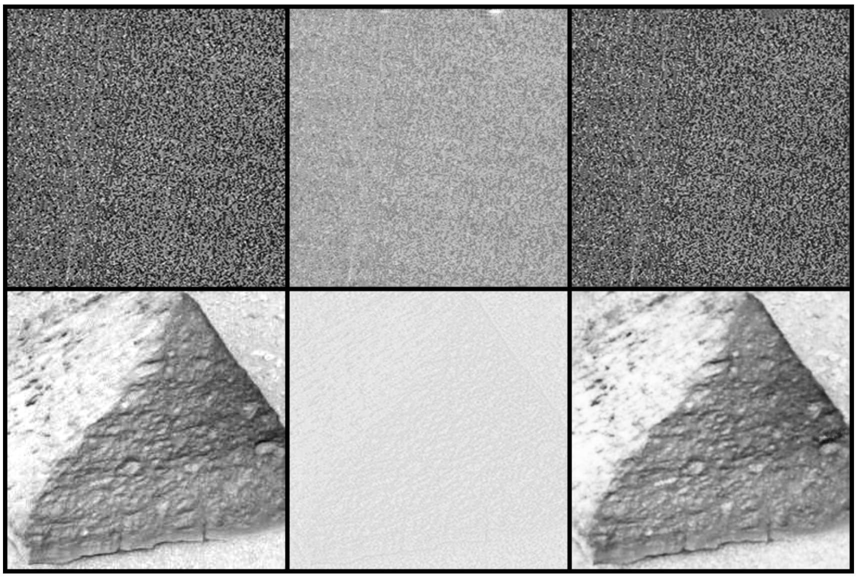

Figure 5.4: Sample translation grid, final epoch 50 checkpoint.

The immediate, epoch-level mechanism is a discriminator shortcut: D_H can reject G_{H2R}'s output purely because it still carries the corrected corpus's genuine sensor grain, without needing to learn what a real rover photograph actually looks like, so G_{H2R} receives a real but semantically uninformative gradient once D_R locks onto that shortcut by epoch 3. This also explains why correcting the corpus resolution made FID *worse* rather than better (Table 5.2): the browse-resolution corpus was smoother and closer to rover-image smoothness, so it did not hand D_R this shortcut as blatantly.

5.2 A Deeper Root Cause: Corpus Diversity

The discriminator-shortcut mechanism above is real, but it is not the whole explanation, and treating it as such would have pointed at the wrong fix. An audit of the corpus that produced the 269.9 result found that of the ten Oxia Planum HiRISE observations the real-resolution pipeline targeted, only seven were available as real, single-band RED JP2 products, and all seven happened to fall in the same region — collapsing the training corpus from the original browse-resolution pipeline's 299 observations across 4 regions down to 7 observations in 1. Separately, Oxia Planum was deliberately selected as the ExoMars landing site *because* it is smooth and low-relief, while the rover-domain training data (Curiosity, Gale Crater) is comparatively rocky — a genuine content mismatch layered on top of the diversity collapse, and one that is somewhat inherent to the problem, since no ground-truth Oxia Planum rover imagery can exist before a rover has actually landed there.

A literature review conducted in response to this finding surfaced a cluster absent from the original three-stage design's reading list: Sat2Density [17], Sat2Scene [12], and CrossViewDiff [11] (Chapter 2, Section 2.4) all translate nadir/satellite imagery into ground-level imagery in Earth remote sensing, and none attempts pure pixel-space translation. Each inserts an explicit geometric or structural intermediate, a density field, a 3D voxel structure, or a separately estimated scene-structure map, between the nadir input and the perspective output, and synthesises texture only once that intermediate has resolved the viewpoint change. Sections 5.3–5.6 report two lines of work that followed from this: a loss-tuned CycleGAN retrain that isolates the discriminator-shortcut question from the corpus-diversity question (Track 1), and the geometry-mediated redesign itself (Track 2 onward), built as described in Chapter 4, Section 4.4.

5.3 Track 1: Isolating the Loss-Weight Question

Reducing λ_{cyc} was one of three candidate fixes the original diagnosis proposed for the discriminator shortcut. Testing it cleanly required a corpus that was not itself confounded by the diversity collapse just diagnosed, so Track 1 deliberately trained on the *original*, browse-resolution corpus (37,054/4,631/4,633 HiRISE patches across the full 4 regions, paired with 20,417/2,552/2,553 rover patches) rather than the smaller, single-region real-resolution one — a resolution/diversity trade-off made explicitly to isolate one variable at a time, and a genuine caveat on how directly Track 1's FID compares against the target, which specifies the true 25 cm/pixel product.

A 15-epoch, 3-way smoke test on this corpus confirmed the hypothesis: $\lambda_{\text{cyc}} = 10$ reproduced the near-identity shortcut, $\lambda_{\text{cyc}} = 1$ broke it but visibly degraded output fidelity, and $\lambda_{\text{cyc}} = 3$ (paired with $\lambda_{\text{idt}} = 1.5$) broke the shortcut cleanly without that degradation. A full 100-epoch run at these validated weights followed (Table 5.3), with no collapse signature at any point (D_R never approached the earlier run's 10^{-6} range).

Table 5.3: Track 1 (loss-tuned) FID trajectory, full 4,633-image test set.

Epoch	FID
15	115.73
25	107.54
35	100.53
45	104.45
55	102.62
100	**95.831**

The run plateaus around FID 100–105 from epoch 35 onward, with the final checkpoint's 95.831 — Track 1's best, and this dissertation's best CycleGAN FID to date — only becoming visible once the full 100 epochs completed; the flat loss curves in the learning-rate decay tail apparently still hid a small real improvement.[1] This is a real fix for the diagnosed shortcut, and a genuine improvement over both the buggy browse-resolution baseline (171.8) and the corrected-but-collapsed corpus (269.9). It is still 1.7× above the target of ≤55, and, more importantly, visual inspection of Track 1's translated outputs at every checkpoint shows the same 2D spatial layout as the input at every epoch — the crater, stripe, and slope positions do not change. This is not a loss-weight problem: a 2D convolutional generator trained under cycle-consistency has no mechanism to reinterpret scene content from a genuinely different physical vantage point, only to restyle it in place. Only an explicit geometric stage — Track 2 onward — can deliver real viewpoint translation, which is the more fundamental reason a pure loss-weight fix cannot reach the target FID on its own.

[1] `epoch_095.pt`'s FID was never computed, having been deleted in a disk-pruning pass before it could be measured; `epoch_100`'s 95.831 stands in as the closest available reading, since the run was in a flat loss plateau from epoch 55 through 100.

5.4 Track 2: Geometry-Mediated Retraining

Track 2 retrains the same architecture and validated loss weights ($\lambda_{\mathrm{cyc}} = 3$, $\lambda_{\mathrm{idt}} = 1.5$) on the geometry-mediated corpus of Chapter 4 (Section 4.4.2): 1,513 rendered ground-view images (1,210/151/152 train/val/test) from 8 real stereo DTMs in the Oxia Planum landing region, in place of a raw HiRISE crop as CycleGAN's domain-A input. 25 epochs completed with no collapse at any point (D_R stayed in a healthy 0.03–0.5 range throughout).

FID against the real rover test set was 426.0 — much worse than Track 1's 95.8, but not a fair comparison: the geometry corpus's 152-image test set is small and visually homogeneous relative to the 2,553-image real rover distribution, and FID was never really the point of this track. Visual inspection was. It confirms geometry-mediation breaks the near-identity shortcut in a way loss-tuning alone (Track 1) cannot: the previously black, unrendered "sky" region above the horizon gets filled with invented texture in the output, and the ground pattern changes structurally rather than merely shifting colour or contrast — the generator is not just copying its input. But output diversity is compressed: three translations from visually distinct inputs (different horizon shapes) share a dominant chevron/zigzag texture motif. Quantified, the ground-region mean pixel difference between outputs (22–28, 0–255 scale) is lower than between their corresponding inputs (29–40) — the generator partially, but not fully, tracks input content, leaning on a shared learned texture template consistent with the corpus's own known risk: only 8 distinct source terrains behind 1,210 training images.

Geometry-mediation is therefore validated as fixing the specific failure mode that broke Tracks 1 and the original run (the architectural inability to translate viewpoint, and the discriminator shortcut respectively), which is a real, positive result for this dissertation's core methodological claim. But the small corpus caps texture diversity in its current form, and Track 2 does not yet produce convincingly varied rover-eye renders on its own, which motivates the separate, paired-supervision texture-synthesis stage (Track 3, Section 5.6) rather than asking the geometry-mediated CycleGAN to solve texture diversity unaided.

5.5 Single-Image DTM Estimation

Track 2's renderer needs a real stereo DTM, and only 8 such products exist for the whole landing region. The single-image DTM estimator of Chapter 4 (Section 4.4.4) is evaluated against 21 held-out real Gale Crater patches with a genuine stereo DTM ground truth, achieving a mean RMSE of **2.02 m** (std. 0.83 m). Qualitatively, rendering the estimator's own output reproduces the same large-scale artefacts a real DTM's render shows at this site, rather than introducing new ones, and it smooths over the real DTM's stereo-interpolation facet noise while tracking coarse structure — an expected and acceptable behaviour for a network predicting a physically smooth quantity.

5.6 End-to-End Chain: Does the Estimator's Error Actually Matter?

The estimator's RMSE is only useful as a number if it can be connected to what actually reaches a viewer: does swapping the real DTM for the estimator's output visibly change the final, texture-synthesised image relative to using ground truth? Answering this required chaining all three geometry-mediation components together for the first time (`run_end_to_end_pipeline.py`, Chapter 4 Section 4.4.6) on the same real Gale HiRISE crop, once against the real stereo DTM and once against the estimator's output, both fed through the same Track 3 ControlNet.

The first two attempts were inconclusive. Run at the corpus's default crop location (the raster centre) and a steep $-15°$ camera pitch, both the real-DTM and estimated-DTM crude renders stayed dominated by a large, flat, low-texture region — a "shelf" artefact — leaving neither condition map with much real geometric signal to propagate. A follow-up run at a shallower $-5°$ pitch, on the assumption that a steep down-look was responsible, produced the same result: the shelf persisted regardless of pitch, ruling out camera angle as the cause.

A diagnostic tool (`find_e2e_crop_candidate.py`) built to investigate scored candidate camera positions across the full DTM by `ground_entropy` (the same metric used to filter the geometry corpus, Chapter 4 Section 4.4.2), and revealed the actual problem: the default crop's render was only 10% sky, yet scored a low ground_entropy of 1.92 — the "shelf" is a large, flat, textureless region of *drawn ground*, not an empty frame, which is exactly why changing pitch never helped. Scanning 150 candidate positions found one scoring 6.08, over 3× higher, at row 6,439/column 2,730 of the same DTM.

Re-running the full chain at this location, at the original $-15°$ pitch, produced the first clean result (Figure 5.6). The real-DTM and estimated-DTM crude renders now visibly differ in horizon and slope shape (Figure 5.5) rather than both being near-blank, and that difference propagates through the Track 3 ControlNet to genuinely different final images: the real-DTM chain's output shows a large dune/rock formation dominating the foreground with strong cast shadows, while the estimated-DTM chain's output shows a flatter rocky plain with different scene framing entirely.

This is the first evidence in the project that the DTM estimator's $\sim$2 m RMSE visibly affects the pipeline's final output, rather than both chains simply converging on the same texture-synthesis prior because the condition signal was too degenerate to carry any real difference.

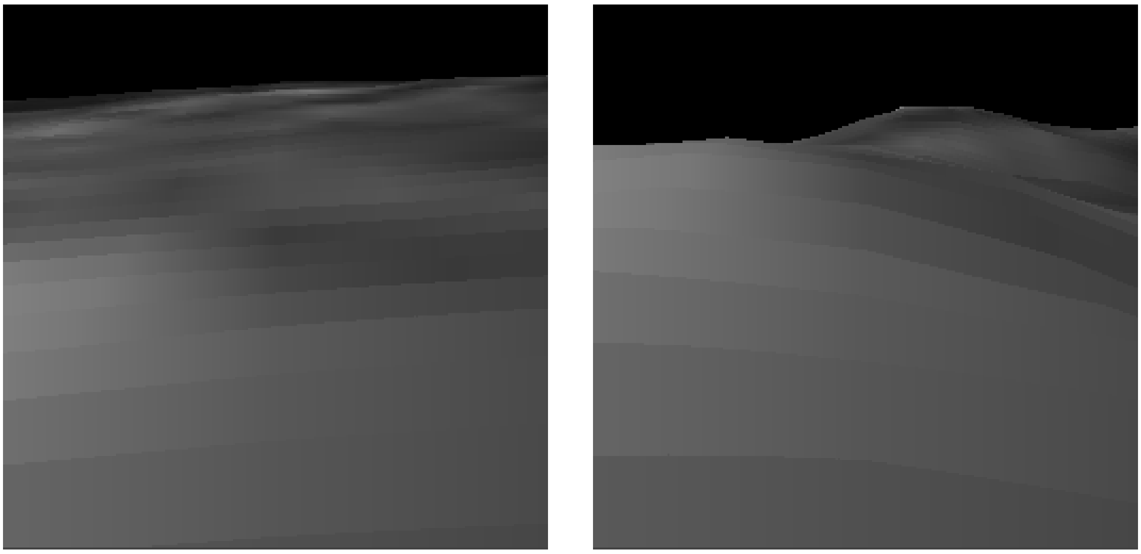

Figure 5.5: Crude perspective renders at the high-structure crop, -15° pitch: real stereo DTM (left) vs. the DTM estimator's own output (right). Unlike every prior test crop, the two now visibly diverge in horizon and slope shape.

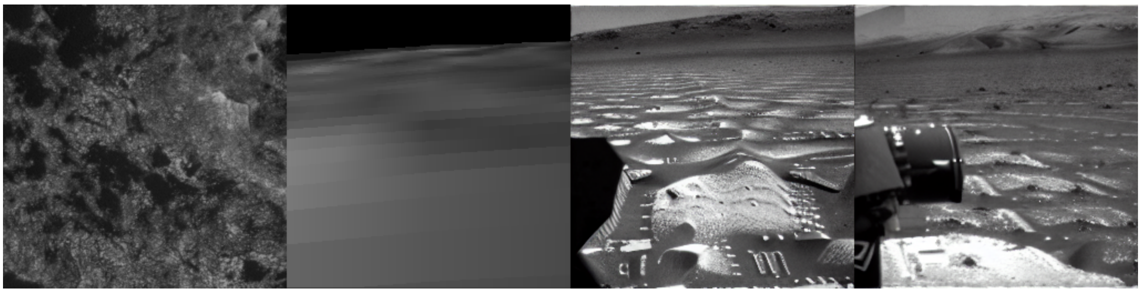

Figure 5.6: End-to-end chain at the high-structure crop, left to right: HiRISE ortho input, real-DTM crude render, real-DTM chain's final Track 3 output, estimated-DTM chain's final Track 3 output. The two final panels diverge in composition, not just texture.

A caveat carries over from Track 3's own limited corpus: both final outputs still show a repeating ripple/dune texture bias relative to a real (loosely, not camera-matched) reference NAVCAM photo at the same site, which shows far more fine rock-clutter chaos. This is consistent with Track 3's training corpus (27 real paired condition/target images) being small, and is a separate, already-understood limitation rather than a new finding from this test.

5.7 Common Evaluation Metrics Across Models

Tracks 1 and 2 were evaluated with FID; Track 3 was evaluated with KID, SSIM, and PSNR instead of FID, because FID's Gaussian-covariance estimate is unreliable at Track 3's small sample size, a documented, deliberate choice (Chapter 4, Section 4.4.6), not an oversight. That left no single number comparable across all three generative stages. KID (an unbiased, small-sample-robust distributional metric) is computed here for Track 1 and Track 2 as well, reusing the same feature-extraction and kernel-MMD^2 implementation already validated for Track 3, so Table 5.4 reports one metric that is valid regardless of each track's very different sample size (4,633 for Track 1, 152 for Track 2, 9–27 for Track 3), alongside a quantitative check of the end-to-end pipeline's two chains that was previously only assessed by eye (Chapter 5, Section 5.6).

Table 5.4: KID across all three generative stages.

Model	KID	n (gen / real)	Notes
Track 1 (loss-tuned CycleGAN)	0.0344	4,633 / 2,553	No geometric mediation
Track 2 (geometry-mediated CycleGAN)	0.4979	152 / 2,553	Breaks the shortcut, compressed diversity
Track 3 (paired ControlNet)	0.192	9 / 9	From Chapter 4's original evaluation

Read alongside FID, KID tells a consistent story: Track 1's translated outputs are, in distributional terms, close to the real rover corpus (KID 0.034), which is unsurprising given Chapter 5, Section 5.3's finding that Track 1 mostly restyles its input in place rather than reinterpreting viewpoint; matching a target distribution's low-level statistics is a substantially easier bar than producing genuine perspective content. Track 2's KID (0.498) is an order of magnitude worse, consistent with the qualitative finding in Section 5.4 that its outputs lean on a shared texture template rather than tracking real rover-photo diversity, and with its much smaller, 152-image evaluation set. Track 3's KID (0.192) sits between the two, computed on the smallest sample of the three (9 images, the same test split Chapter 4 originally evaluated); its SSIM (0.186) and PSNR (11.4 dB) values, reported alongside KID for that track, measure a stricter, different question, exact reconstruction of one specific real photo, which a model that hallucinates plausible but unseen rock and ripple detail is not expected to score well on regardless of how convincing its output looks.

Table 5.5 extends this to the end-to-end pipeline (Section 5.6), turning its previously qualitative comparison into numbers. The real-DTM and estimated-DTM chains' final outputs differ substantially from each other (SSIM 0.269, LPIPS 0.470), which is the quantitative form of the claim that the DTM estimator's error visibly propagates to the final image. Both chains score similarly low against the loose, not camera-matched, real reference photograph (LPIPS 0.537 and 0.540 respectively); this pair of numbers is not a meaningful ranking between the two chains, since neither render was ever expected to reconstruct that specific, non-corresponding photograph, but their near-equality does confirm that the real-vs-estimated-DTM difference above reflects the two chains actually differing from each other, not one of them coincidentally drifting closer to an unrelated reference image.

Table 5.5: Quantitative comparison of the end-to-end pipeline's two chains (Chapter 5, Section 5.6), 256×256 grayscale-normalised, matching Track 3's evaluation preprocessing.

Comparison	SSIM	PSNR (dB)	LPIPS
Real-DTM chain vs. estimated-DTM chain	0.269	11.94	0.470
Real-DTM chain vs. loose reference photo	0.083	10.35	0.537
Estimated-DTM chain vs. loose reference photo	0.109	10.03	0.540

Read against Objective O2 (Section 1.2), this chapter reports genuine progress and genuine open gaps in roughly equal measure. On the positive side: the original failure was traced to a specific, evidenced mechanism at two levels (a discriminator shortcut, and the corpus diversity collapse behind it), the loss-weight fix for the first was validated in isolation (Track 1), geometry-mediation was validated as fixing the second in the way loss-tuning cannot (Track 2), and today's high-structure-crop test is the first real evidence that the geometry-mediated chain's weakest link — the single-image DTM estimator standing in for a real stereo DTM — actually matters to what a viewer would see, rather than being masked by an uninformative test.

On the open side: FID remains 1.7× above target even at Track 1's best; Track 2's output diversity is compressed by its small, 8-terrain corpus; and Track 3's texture synthesis is trained on only 27 real pairs and shows a corresponding texture bias. The architecture that produced these results, a deterministic renderer for geometry, a single-image estimator standing in for stereo coverage, and a separately-trained ControlNet for texture, is a different family from the physics-constrained Neural Schrödinger Bridge Chapter 3 originally specified; Chapter 3, Section 3.2.3

states why the design changed, and Section 5.8 below reports every result in this chapter against the original target table (Chapter 3, Table 3.2) directly, rather than leaving the comparison implicit. Chapter 6 addresses the remaining open items as concrete next steps.

5.8 Results Against the Original Targets

Table 5.6 restates every quantitative target Chapter 3 set (Table 3.1, Table 3.2, and Section 3.5's six-metric evaluation framework) against what this project actually measured, so the gap is a single table rather than something a reader has to reconstruct from six sections. A dash marks a target that was not evaluated at all, not a target that was evaluated and missed; Section 6.5 and Chapter 6 explain which of those were cut deliberately, for stated reasons, versus left short by time.

Table 5.6: Every quantitative target from Chapter 3 against what was actually measured.

Target (Chapter 3)	Original goal	What was measured	Met?
FID (view translation)	$\leq$55	95.831 (Track 1, best CycleGAN checkpoint)	No
KID (view translation, all tracks)	not specified	see Section 5.7	Partial
Shadow-angle error	$< 10°$	not evaluated (Section 6.5)	—
Hapke BRDF residual	minimised	not evaluated; auxiliary network never built (Chapter 3, Section 3.2.3)	—
Hallucination rate	$< 5\%$	not evaluated; auxiliary classifier never built	—
VR render throughput	>72 fps	not evaluated; Stage 3 out of scope (Section 6.4)	—
HiRISE corpus size	50,000 patches	37,054 (Track 1); 1,513 geometry-render patches (Track 2)	Partial
Rover corpus size	30,000 images	20,417 (unpaired, Tracks 1–2); 27 real pairs (Track 3)	Partial
DTM estimator accuracy	not specified	RMSE 2.02 m, 21 Gale patches (cf. 1.05 m [21])	New metric

Read plainly, the project met none of Chapter 3's original numeric thresholds. Chapter 7's assessment against the original objectives argues that this table understates what was actually learned, since three of the four unmet metrics were never attempted because their prerequisite auxiliary networks were scoped out for stated reasons (Chapter 3, Section 3.2.3), not because they were attempted and failed, and the FID gap has a specific, evidenced architectural cause (Section 5.3) rather than an unexplained shortfall. Both readings are reported here rather than only the more favourable one.

6 SCOPE DECISIONS AND FUTURE WORK

The previous draft of this chapter posed three open forks rather than resolving them, on the reasoning that each was a decision for the author rather than something to settle unilaterally while editing prose. This chapter records how each was resolved, what that resolution means for the submitted dissertation's scope, and what remains as genuine future work beyond it.

6.1 The Architecture Reconciliation

Chapter 3, Section 3.2.3 now states in full why the geometry-mediated pipeline (renderer, DTM estimator, ControlNet) was adopted in place of the physics-constrained Neural Schrödinger Bridge originally specified. In brief: the corpus-diversity diagnosis (Chapter 5, Section 5.2) showed that an explicit geometric intermediate is needed regardless of which learned architecture sits either side of it, and once the pose-pairing pipeline (Chapter 4, Section 4.4.5) produced real paired data, a paired conditional-diffusion objective became a more direct fit than an unpaired bridge built for exactly the situation, no available correspondence, that no longer applied. Implementing UNSB as originally specified remains possible future work, and would now start from a stronger position than Chapter 3 anticipated: a validated geometric intermediate and a real, if small, paired corpus, rather than nothing. It is not pursued in this dissertation because the two findings above already answer the question UNSB was chosen to answer, and because its three auxiliary physics networks (shadow-direction, Hapke inversion, geological classifier) remain unbuilt, unscoped, and orders of magnitude larger a task than the time remaining before submission allows.

6.2 Track 2's Corpus-Diversity Limitation

Track 2 (Chapter 5, Section 5.4) validated that geometry-mediation breaks the near-identity shortcut, but its output diversity is compressed by an 8-terrain source corpus, all the stereo DTM coverage that exists for the landing region at the strict bounding box used. This is reported as an accepted, documented limitation rather than something to fix before submission: the two remaining alternatives, widening the DTM search to include neighbouring, geologically different terrain, or replacing CycleGAN's restyling objective with something more resistant to the shared-texture-template failure, would each reopen a design question (content mismatch, or architecture choice) already settled elsewhere in this dissertation for good reason, and neither is achievable to a validated standard in the time remaining. The shortcut-breaking result, which is the finding Track 2 was built to test, stands on its own regardless of this limitation.

6.3 Stage 1 (DiffusionSR): Formally Out of Scope

The CTX/HiRISE co-registration pipeline for Stage 1 (nadir super-resolution) reached a five-image smoke test and was not developed further. It is formally scoped out of this dissertation: Chapter 3, Section 3.1 already notes that Stage 1 is optional wherever HiRISE coverage exists natively, which is true at every site this project actually used (Oxia Planum, Gale Crater), and the geometry-mediated pipeline that Objective O2 depends on was built and evaluated entirely on that assumption. Extending coverage to sites without native HiRISE resolution via Stage 1 remains valid future work, but was not a prerequisite for any result in Chapter 5.

6.4 Stage 3 (Gaussian Splatting): Formally Out of Scope

No code exists for the 3D Gaussian Splatting rendering stage Chapter 3, Section 3.1 specifies as Stage 3, and Objective O3's $\geq$72 fps VR throughput target was not assessed. This is formally scoped out of the submitted dissertation. Fitting a 3DGS scene needs multiple, multi-view-consistent perspective renders of the same location; everything Chapter 5 evaluates is single-image output from a fixed or independently sampled camera pose, so scoping Stage 3 out did not cost this dissertation any result it would otherwise have reported, and using the remaining time for the evaluation extension below and the Conclusions chapter (Chapter 7) instead was judged the better use of it.

6.5 Common Evaluation Metrics Across Models

Chapter 3's evaluation framework (Section 3.5) specifies six measurements; before this chapter, only FID and the DTM estimator's RMSE (not itself part of the original six) had been computed. Two of the remaining four, Hapke BRDF residual and hallucination rate, are gated on the auxiliary networks Section 6.1 scopes out, and are not pursued here for the same reason. Shadow-angle error is technically the cheapest of the four to add, since HiRISE products already carry solar geometry metadata and the renderer already produces a geometrically exact reference to compare against, but estimating a generated image's own apparent shadow direction still requires a small estimator that does not yet exist; it is left as future work rather than attempted at low confidence this close to submission.

What this dissertation adds instead is a common metric across the three generative stages that Chapter 3's own framework did not anticipate needing: Track 1 and Track 2 originally reported only FID, while Track 3 reported KID instead of FID specifically because FID's Gaussian-covariance estimate is unreliable at Track 3's small sample size (Chapter 4, Section 4.4.6). KID (an unbiased polynomial-kernel MMD^2 estimator, well-defined regardless of sample size) is computed for Track 1 and Track 2 as well, giving one distributional metric that is valid across all three stages at once, alongside each track's original metric for continuity with Chapter 3's literature comparison table. Section 5.7 in Chapter 5 reports the results.

6.6 Indicative Priorities for Extending This Work

With the three scope forks above resolved and the common-metrics extension complete, what remains as genuine future work, in priority order, is: implementing UNSB and its physics losses on the now-validated geometric intermediate (Section 6.1), which would let the dissertation's original architecture be assessed on equal footing with what was actually submitted; training the physics-informed ControlNet variant designed but not built (Chapter 4, Section 4.4.7), which targets Track 3's specific chevron-texture artefact directly; adding shadow-angle error once a lightweight estimator exists; scaling the Track 2 and Track 3 corpora, most directly by widening Gale Crater's pose-pairing search beyond its current DTM coverage, since that pipeline (unlike Oxia Planum's 8-DTM ceiling) is not obviously saturated; and, only once Stage 2's output is multi-view-consistent, Stage 3 Gaussian Splatting and Stage 1 super-resolution for sites without native HiRISE coverage.

7 CONCLUSIONS

7.1 Assessment Against the Objectives

Section 1.2 set out four objectives. Judged against them directly:

O1 (nadir super-resolution from coarser CTX/CaSSIS inputs) was not achieved. The co-registration pipeline it depends on reached a five-image smoke test and was formally scoped out (Chapter 6, Section 6.3) once it became clear that neither test site used in this project needed it: HiRISE covers Oxia Planum and Gale Crater natively.

O2 (unpaired nadir-to-perspective translation, physically consistent) was achieved in a different architectural form from the one proposed. A viewpoint-translating, texture-synthesising pipeline was built and evaluated end-to-end, and Chapter 5's results show it working: geometry-mediation demonstrably breaks the failure mode that blocked pure pixel-space translation, and the end-to-end chain shows the DTM estimator's error visibly reaching the final image rather than being absorbed unnoticed. Chapter 3, Section 3.2.3 argues that this substitution was the correct response to what the project's own evidence showed, not a lowering of ambition; Chapter 5, Section 5.8 reports plainly that its FID (95.8) still falls short of the original target ($\leq$55), by a margin traced to a specific, understood architectural limit of restyling-based translation rather than an unexplained shortfall.

O3 (real-time 3D Gaussian Splatting rendering at $\geq$72 fps) was not achieved and was formally scoped out (Chapter 6, Section 6.4). No code exists for it, and no claim is made otherwise.

O4 (quantified fidelity and physical plausibility) was achieved partially. FID, KID, RMSE, SSIM, and PSNR were computed, and Chapter 5 reports every one of them, including where they fall short of target. Shadow-angle error, Hapke BRDF residual, and hallucination rate, the three metrics most directly tied to physical plausibility rather than perceptual similarity, were not computed, because two of the three depend on auxiliary networks this project did not build (Chapter 3, Section 3.2.3) and the third was judged too large an addition this close to submission (Chapter 6, Section 6.5).

Two of four objectives were met in substance, one in a revised form with an explicit rationale for the revision, and two were formally and explicitly not attempted. That is a narrower final scope than the original proposal, arrived at deliberately rather than by running out of time on an unstated plan.

7.2 Reflection on Project Management

The original plan assumed HiRISE and rover corpora that, once acquired, would train directly; in practice, two separate corpus-quality problems (a browse-resolution substitution bug, and a severe diversity collapse from an overly narrow fetch default) consumed time that the plan had allocated to Stage 2 architecture work, and were only found because CycleGAN's training behaviour was diagnosed rather than accepted at face value. That diagnosis, not any part of the original Gantt-level plan, is what actually drove the project's direction from roughly its midpoint onward: the geometry-mediated redesign exists because a specific, evidenced failure pointed at a specific cause, not because it was scheduled.

This is a genuine deviation from the original plan's assumption that Stage 2 architecture work could begin once corpora were acquired. In hindsight, the corpus-diversity risk (a single fetch script defaulting to one region) was not one the original design or its risk assessment anticipated, and a corpus audit earlier in the timeline, before any training run, would have caught it without the cost of a full 50-epoch training run that produced an uninformative result. That audit habit was adopted for every corpus built afterward (the geometry corpus, the DTM estimator's training set, the pose-matched pairing corpus), each of which was checked for size, diversity, and degenerate content before being trained on, which is the direct, applied lesson from the earlier miss.

Architecture and infrastructure decisions were made incrementally, each justified by evidence from the previous step (loss-weight tuning isolated the discriminator shortcut; geometry-mediation isolated the viewpoint-translation limit; real pose-pairing removed the premise for an unpaired-only architecture), rather than committed to upfront. This is a defensible way to manage a project whose central technical risk, whether any translation architecture could close a domain gap this large, was explicitly flagged as unresolved in the literature review (Chapter 2) before implementation began. It did mean deferring the reconciliation between the approved methodology and the system actually being built for longer than it should have been left implicit; Chapter 6 records that reconciliation now, in the project's own words, rather than leaving two chapters describing different systems.

7.3 Summary

This dissertation set out to close the nadir-to-perspective domain gap for Mars landing-site imagery using generative modelling, and to render the result for VR. What it delivers is a working, evaluated, geometry-mediated translation pipeline for the middle of that goal, real evidence for why that architecture replaced the one originally proposed, and an explicit account of what was cut and why, rather than left unstated. The FID gap against the original target, the small size of the real paired corpus, and the unbuilt physical-plausibility metrics are reported as open limitations, not hidden ones. Chapter 6 sets out the specific next steps that follow from each: implementing the physics-constrained bridge on top of the now-validated geometric intermediate, training the physics-informed ControlNet variant already designed, and extending the corpus and evaluation coverage this dissertation's own scope decisions left for future work.

BIBLIOGRAPHY

[1] Aithal, S. K. , Maini, P. , Lipton, Z. C. , and Kolter, J. Z. . Understanding hallucinations in diffusion models through mode interpolation. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, 2024.

[2] Cao, J. , Huang, R. , Ye, Z. , Xu, Y. , and Tong, X. . Generative 3D reconstruction of martian surfaces using monocular images. In *ISPRS Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, volume XLVIII-3-2024, pages 51–56, 2024. doi: 10.5194/isprs-archives-XLVIII-3-2024-51-2024.

[3] Catalano, G. L. and Soccini, A. M. . Inpainting the red planet: Diffusion models for the reconstruction of martian environments in virtual reality. *arXiv preprint arXiv:2510.14765*, 2025.

[4] Almeida, M. , de, Brydon, G. , Persaud, D. M. , Williamson, J. H. , and Henderson, P. . Neural 3D reconstruction of planetary surfaces from descent-phase wide-angle imagery. *arXiv preprint arXiv:2604.13235*, 2025.

[5] Giusti, L. , Garcia, J. , Cozine, S. , Suen, D. , Nguyen, C. , and Alimo, R. . MaRF: Representing mars as neural radiance fields. *arXiv preprint arXiv:2212.01672*, 2022.

[6] Hapke, B. . *Theory of Reflectance and Emittance Spectroscopy*. Cambridge University Press, Cambridge, UK, 2nd edition, 2012. doi: 10.1017/CBO9781139025683.

[7] Kerbl, B. , Kopanas, G. , Leimkühler, T. , and Drettakis, G. . 3D Gaussian splatting for real-time radiance field rendering. In *ACM SIGGRAPH 2023 Conference Proceedings*. ACM, 2023. doi: 10.1145/3588432.3591563.

[8] Khanna, S. , Liu, P. , Zhou, L. , Meng, C. , Rombach, R. , Burke, M. , Lobell, D. B. , and Ermon, S. . DiffusionSat: A generative foundation model for satellite imagery. In *International Conference on Learning Representations (ICLR)*, 2024.

[9] Kim, B. , Kwon, G. , Kim, K. , and Ye, J. C. . Unpaired image-to-image translation via neural Schrödinger bridge. In *International Conference on Learning Representations (ICLR)*, 2024.

[10] Li, L. , Fan, Z. , Cong, W. , Liu, X. , Yin, Y. , Foutter, M. , Pan, P. , You, C. , Wang, Y. , Wang, Z. , Zhao, Y. , Pavone, M. , and Wei, Y. . Martian world model: Controllable video synthesis with physically accurate 3D reconstructions. *arXiv preprint arXiv:2507.07978*, 2025.

[11] Li, W. , He, J. , Ye, J. , Zhong, H. , Zheng, Z. , Huang, Z. , Lin, D. , and He, C. . CrossViewDiff: A cross-view diffusion model for satellite-to-street view synthesis. *arXiv preprint arXiv:2408.14765*, 2024.

[12] Li, Z. , Li, Z. , Cui, Z. , Pollefeys, M. , and Oswald, M. R. . Sat2Scene: 3D urban scene generation from satellite images with diffusion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7141–7150, 2024.

[13] Lu, C. and Su, C. . Super resolution reconstruction of mars thermal infrared remote sensing images integrating multi-source data. *Remote Sensing*, 17(13):2115, 2025. doi: 10.3390/rs17132115.

[14] Martínez-Petersen, A. , Gerdes, L. , Rodríguez-Martínez, D. , and Pulgar, C. J. , Pérez-del. High-fidelity 3D reconstruction for planetary exploration. *arXiv preprint arXiv:2602.13909*, 2025.

[15] Osadnik, R. , Gómez, P. , Bohacek, E. , and Bahia, R. . MCTED: A machine-learning-ready dataset for digital elevation model generation from mars imagery. *arXiv preprint arXiv:2509.08027*, 2025.

[16] Paul, A. , Deevi, S. A. , Chaudhary, A. , Lagisetty, R. K. , Anoop G. L., and Ganesh Kumar R. Neural radiance fields in space applications: A comprehensive review. *International Journal of Advanced Research*, 13(3):525–559, 2025. doi: 10.21474/IJAR01/20596.

[17] Qian, M. , Xiong, J. , Xia, G.-S. , and Xue, N. . Sat2Density: Faithful density learning from satellite-ground image pairs. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 3683–3692, 2023.

[18] Rathkopf, C. . From hallucination to reliability: Generative modeling and the structure of scientific inference. *arXiv preprint arXiv:2504.08526*, 2025.

[19] Serrano, N. , McGuire, P. , Mayer, D. , Huertas, A. , and Arvidson, R. . Predicting HiRISE-equivalent rock density on mars using CTX image features. In *AIAA SPACE 2013 Conference and Exposition*. American Institute of Aeronautics and Astronautics, 2013.

[20] Tao, Y. , Conway, S. J. , Muller, J.-P. , Putri, A. R. D. , Thomas, N. , and Cremonese, G. . Single image super-resolution restoration of TGO CaSSIS colour images: Demonstration with perseverance rover landing site and mars science targets. *Remote Sensing*, 13(9):1777, 2021. doi: 10.3390/rs13091777.

[21] Tao, Y. , Muller, J.-P. , Xiong, S. , and Conway, S. J. . MADNet 2.0: Pixel-scale topography retrieval from single-view orbital imagery of mars using deep learning. *Remote Sensing*, 13 (21):4220, 2021. doi: 10.3390/rs13214220.

[22] Wu, F. , Jiang, X. , Fu, T. , Fu, Y. , Xu, D. , and Zhao, C. . RSTSRN: Recursive swin transformer super-resolution network for mars images. *Applied Sciences*, 14(20):9286, 2024. doi: 10.3390/app14209286.

[23] Xiao, B. , Wang, P. , He, Q. , and Dong, M. . Deterministic image-to-image translation via denoising Brownian bridge models with dual approximators. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

[24] Zhu, C. , Liu, Y. , Huang, S. , and Wang, F. . Taming a diffusion model to revitalize remote sensing image super-resolution. *Remote Sensing*, 17(8):1348, 2025. doi: 10.3390/rs17081348.

[25] Zou, Z. , Zhou, Y. , Li, C. , and Dong, Z. . PFMGAN: A generative adversarial network with physics fusion for DTM generation from martian monocular images. *Remote Sensing*, 18(9): 1354, 2026. doi: 10.3390/rs18091354.

8 REPRODUCIBILITY DETAILS

This appendix collects the hyperparameters, infrastructure, and evaluation-artefact locations referenced throughout Chapters 4–5, in one place, for a reader wanting to reproduce or extend a specific run rather than re-derive its settings from the surrounding narrative.

8.1 Training Hyperparameters

Table 8.1: Hyperparameters for every trained model reported in Chapter 5.

	Track 1	Track 2	DTM estimator	Track 3 ControlNet
Base architecture	CycleGAN (ResNet-9 gen.)	CycleGAN (ResNet-9 gen.)	ControlNet+LoRA	ControlNet (SD1.5)
Training pairs/images	37,054 / 20,417 (unpaired)	1,210 (unpaired vs. rover)	Gale DTM/ortho patches	21 real pairs
Epochs / steps	100 epochs	25 epochs	20,000 steps	50 epochs
λ_{cyc} / λ_{idt}	3.0 / 1.5	3.0 / 1.5	—	—
Batch size	4	4	4	2
Learning rate	2×10^{-4}	2×10^{-4}	1×10^{-5}	1×10^{-5}
LoRA rank	—	—	16	—
Hardware	RTX A4000	A100 (Eureka2)	A100 (Eureka2)	A100 (Eureka2)
Wall-clock time	˜52 hours	˜16 minutes	—	—

Track 1 and Track 2 share their loss-weight configuration ($\lambda_{\text{cyc}} = 3.0$, $\lambda_{\text{idt}} = 1.5$), validated by the three-way smoke test reported in Chapter 5, Section 5.3; the same A100 that completed Track 2's 25 epochs in roughly sixteen minutes took the A4000's RTX A4000 approximately fifty-two hours for Track 1's 100, the empirical basis for this project's general preference for the A4000 on short, queue-free jobs and Eureka2's SLURM cluster on GPU-bound ones, noted throughout Chapter 4.

8.2 Evaluation Artefacts

Every quantitative result in Chapter 5 is backed by a JSON file recording the metric value, sample sizes, checkpoint identifier, and a timestamp, rather than a number transcribed once and left unverifiable:

- `CycleGAN/Data/eval/ fid_results.json` — Track 1 FID.
- `CycleGAN/Data/eval/ kid_results_track1.json`, `kid_results_track2.json` — Table 5.4's KID values.
- `ControlNet/Data/eval/ mars_model_eval.json` — Track 3 KID/SSIM/PSNR.
- `ControlNet/Data/eval/ dtm_estimator_eval.json` — DTM estimator RMSE.
- `ControlNet/Data/eval/ e2e_pipeline_metrics.json` — Table 5.5's end-to-end SSIM/PSNR/LPIPS values.

8.3 Infrastructure

Training and evaluation ran across three machines with distinct roles: a local development machine (no CUDA, used for data pipeline development, testing, and disk-heavy fetch steps such as `prepare_e2e_crop.py`), an RTX A4000 workstation (25 GB NFS home quota, no SLURM queue, used for short or disk-light GPU jobs), and Surrey's Eureka2 SLURM cluster (A100 80 GB nodes, used for larger training runs and jobs where queue wait time is acceptable). Checkpoints and generated-image sets were treated as ephemeral and regenerable from a checkpoint plus a fixed evaluation script where disk quota required pruning them, rather than a project asset to be preserved indefinitely; every pruning decision that removed an unevaluated checkpoint is noted at the point in Chapter 5 where its absence matters.